Dr Michelle Lussier

THE THORACIC SPINE

TARGET CONCEPTS

- Define thoracic spine and costal cage anatomy
- Understand thoracic intervertebral segment biomechanics
- Explore thoracic spine and costal cage dysfunction
- Describe examination and evaluation procedures
- Prescribe evidence-informed interventions

TALKING POINTS

- Overview
- Functional anatomy
- Pathophysiology
- Evaluation
- Management

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WHAT DO WE KNOW ABOUT THE THORACIC SPINE RIGHT NOW?

EPIDEMIOLOGY

- Degenerative disc disease occurs as commonly as in the cervical and lumbar spines, however, signs and symptoms are less often made manifest
- Only 2% of thoracic spine sign and symptoms are of discogenic origin

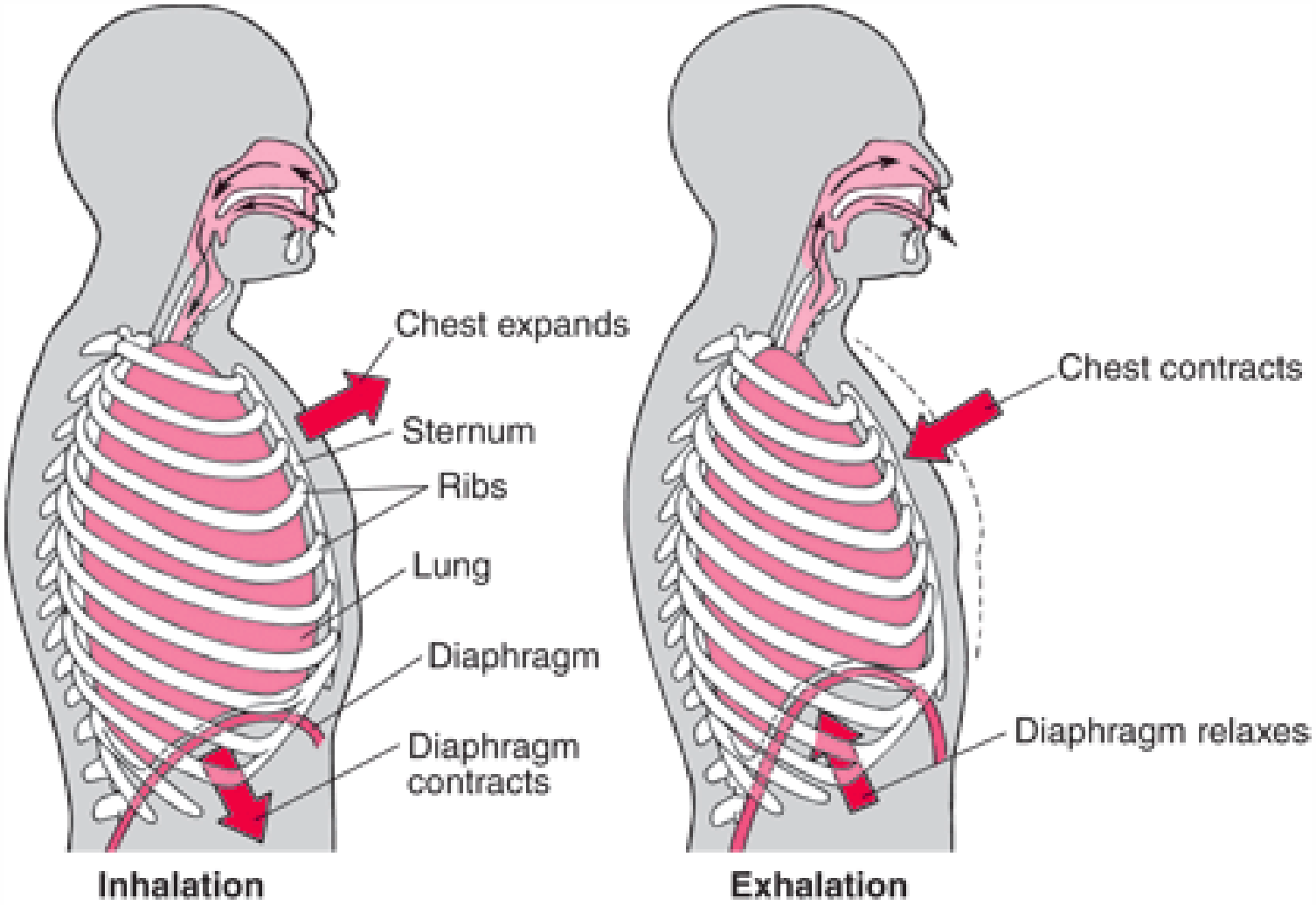
THORACIC SPINE FUNCTION

- Protection
- Respiration
- Postural stability
- Rotational mobility
- Load transfer
- Transmission of power



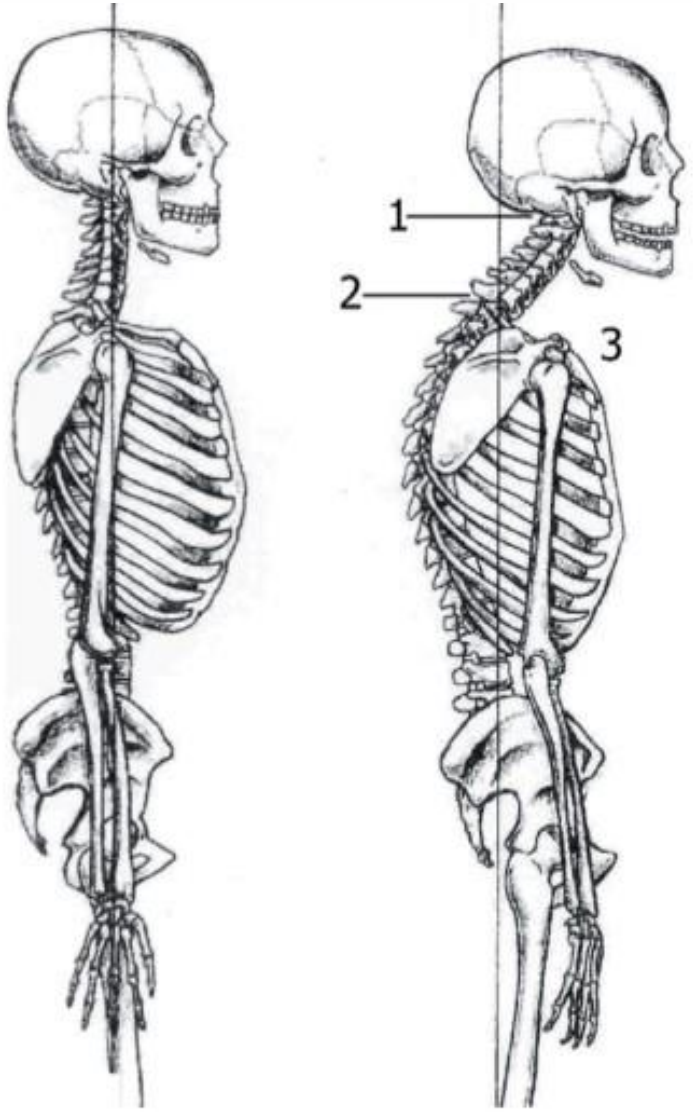
Gene Schiavone

RESPIRATION



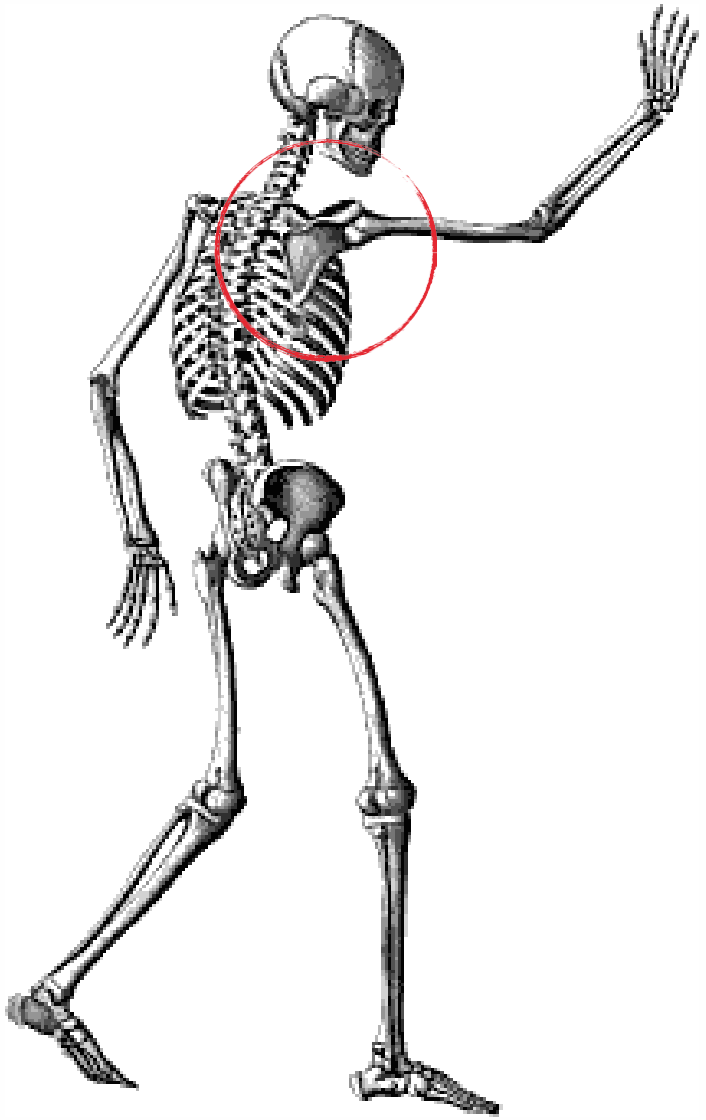
POSTURAL STABILITY FOR THE HEAD AND NECK

- Thoracic hyperkyphosis dictates forward head posturing to maintain a level gaze
- Habitual forward head posture is the leading cause of upper thoracic spine hypomobility



POSTURAL STABILITY FOR THE SHOULDER GIRDLE

- Alteration of typical thoracic kyphosis adversely affects resting scapular alignment
- The upper thoracic spine and the first six ribs must extend and rotate in order to achieve full upper extremity elevation



ROTATIONAL MOBILITY

- Rotational hypomobility of the thorax predisposes the cervical and lumbar spine to segmental deterioration
- Fryette's Principle III: When motion is introduced in one plane it will modify motion in the other two planes





FUNCTION

LOAD TRANSFER: PUSHING



FUNCTION

LOAD TRANSFER: PULLING



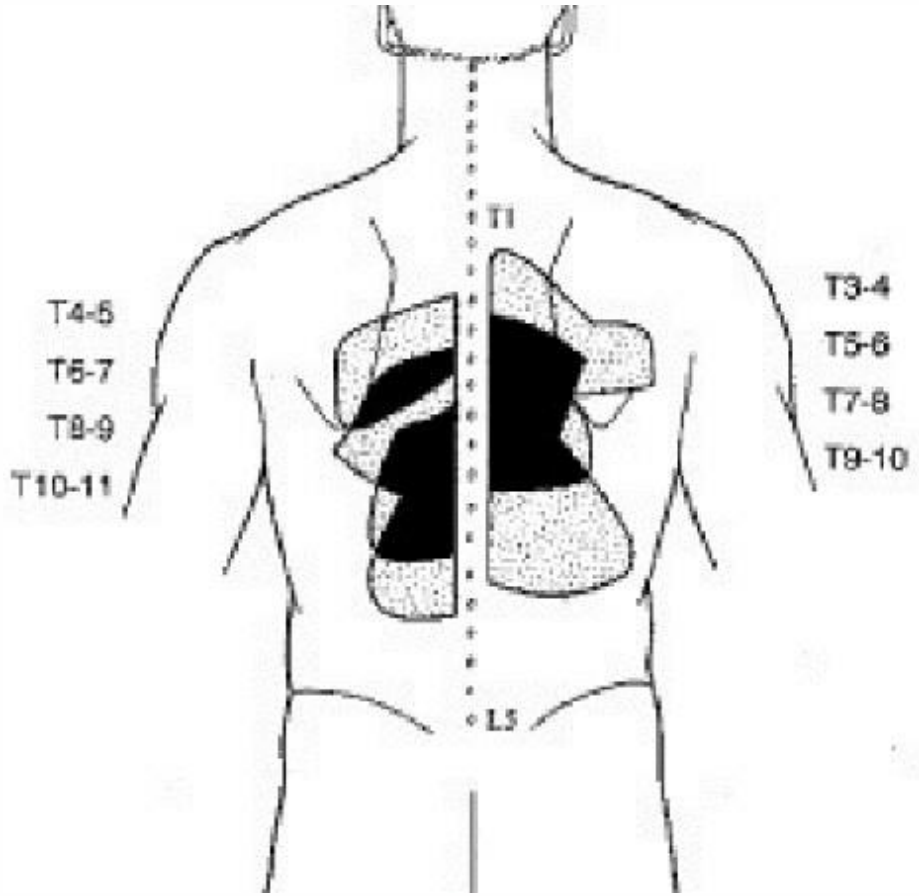
FUNCTION

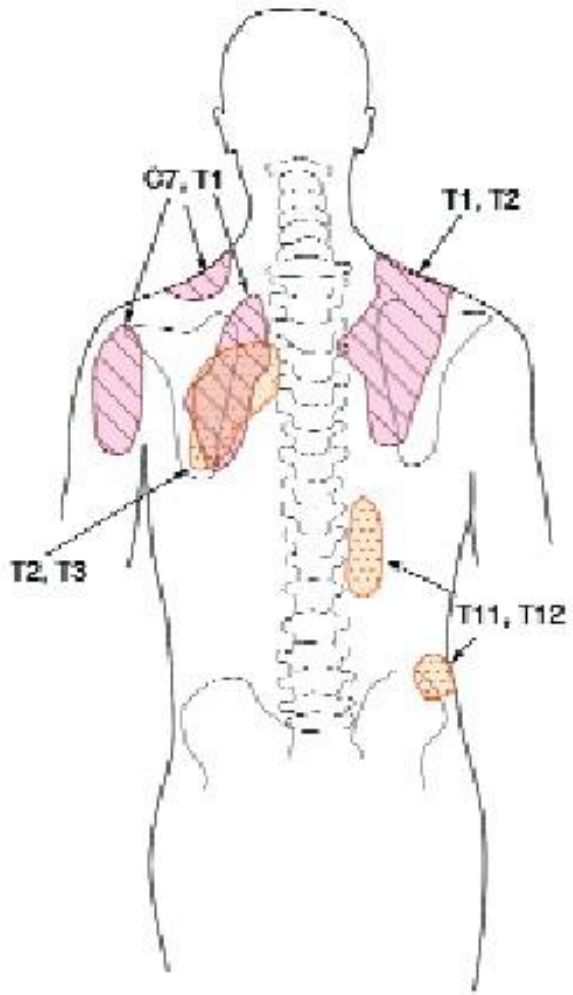
TRANSMISSION OF POWER

TALKING POINTS

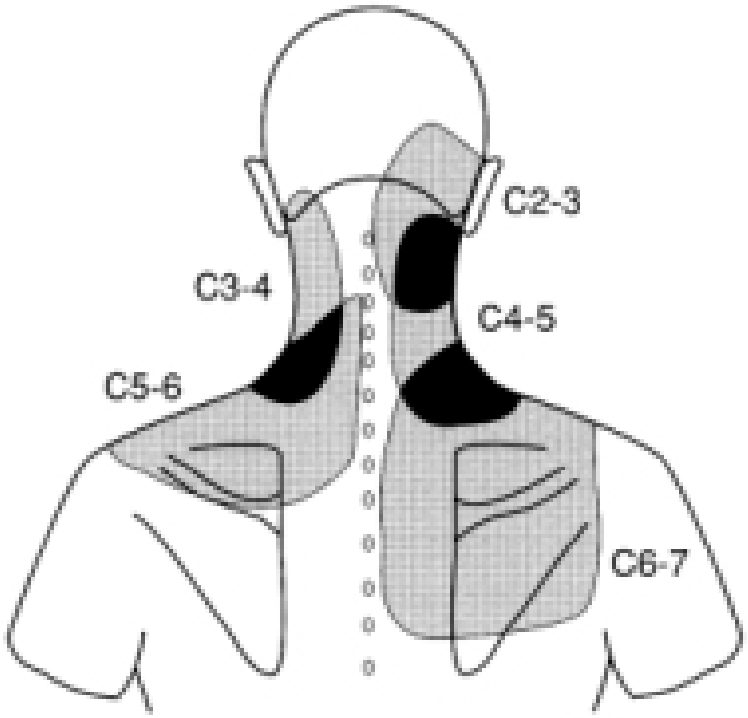
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THORACIC SPINE REFERRAL PATTERNS



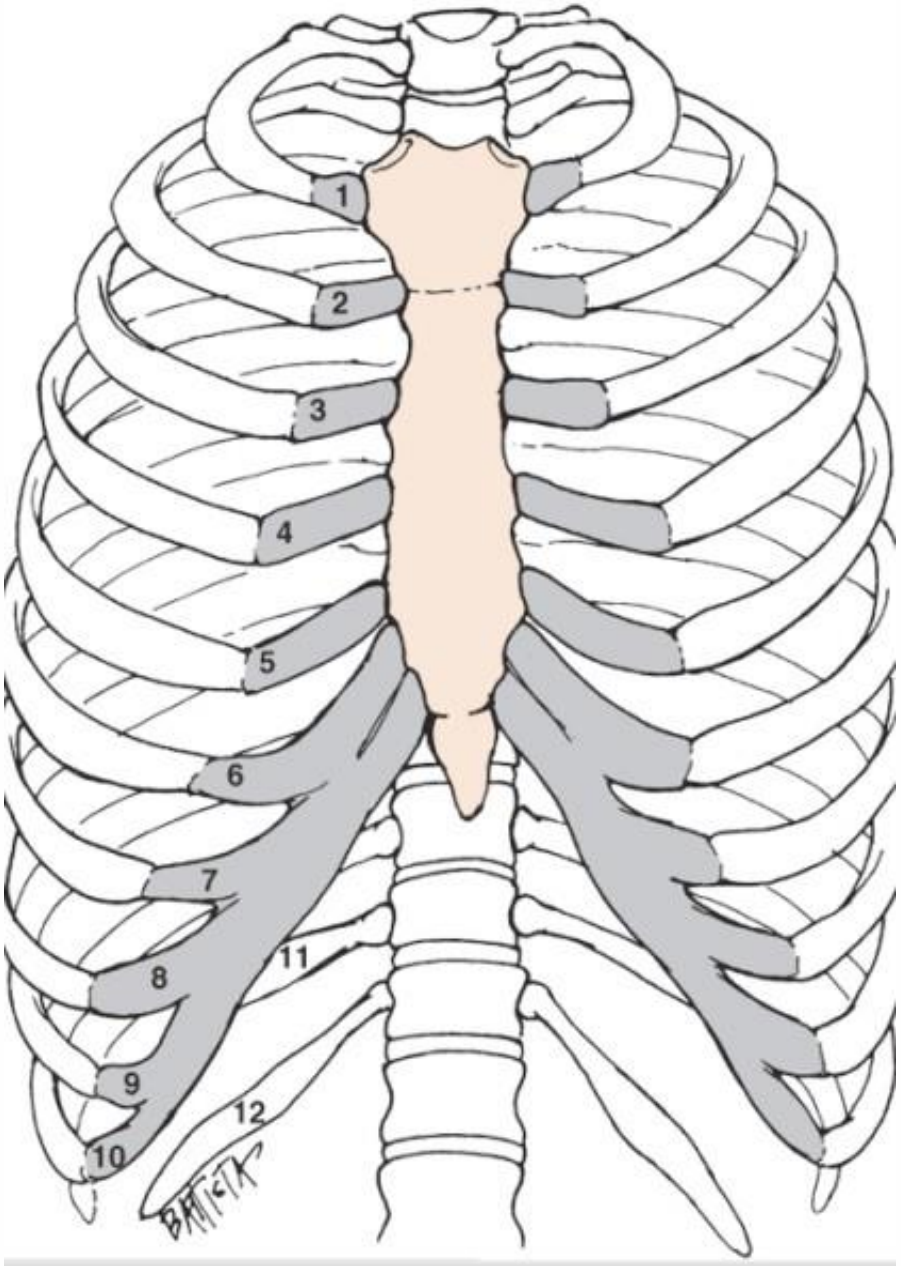


CERVICAL SPINE REFERRAL PATTERNS

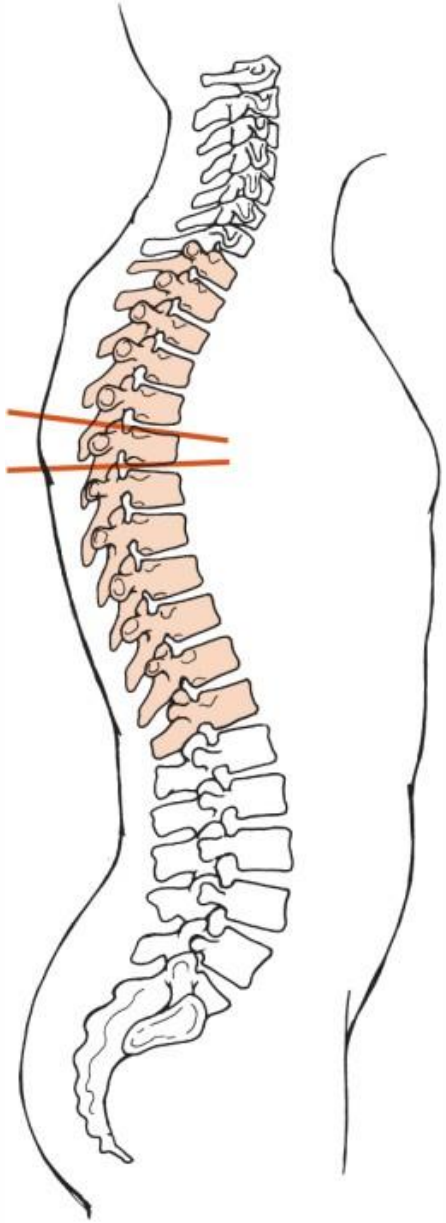


STRUCTURAL ORGANIZATION

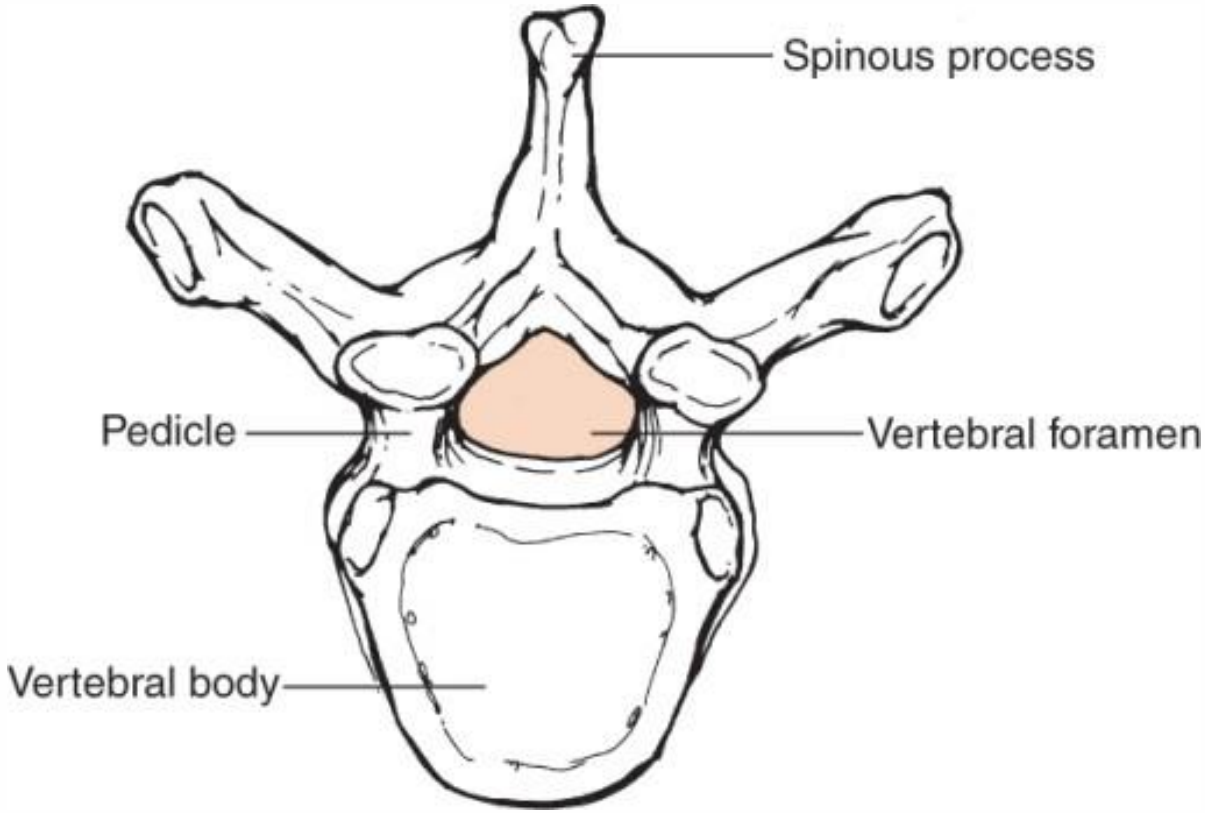
- Vertebromanubrial
  - T1-2, Atypical ribs 1-2
- Vertebrosternal
  - T3-7, True ribs 3-7
- Vertebrochondral
  - T8-10, Mobile ribs 8-10
- Thoracolumbar
  - T11-12, Floating ribs 11-12



STRUCTURAL ORGANIZATION

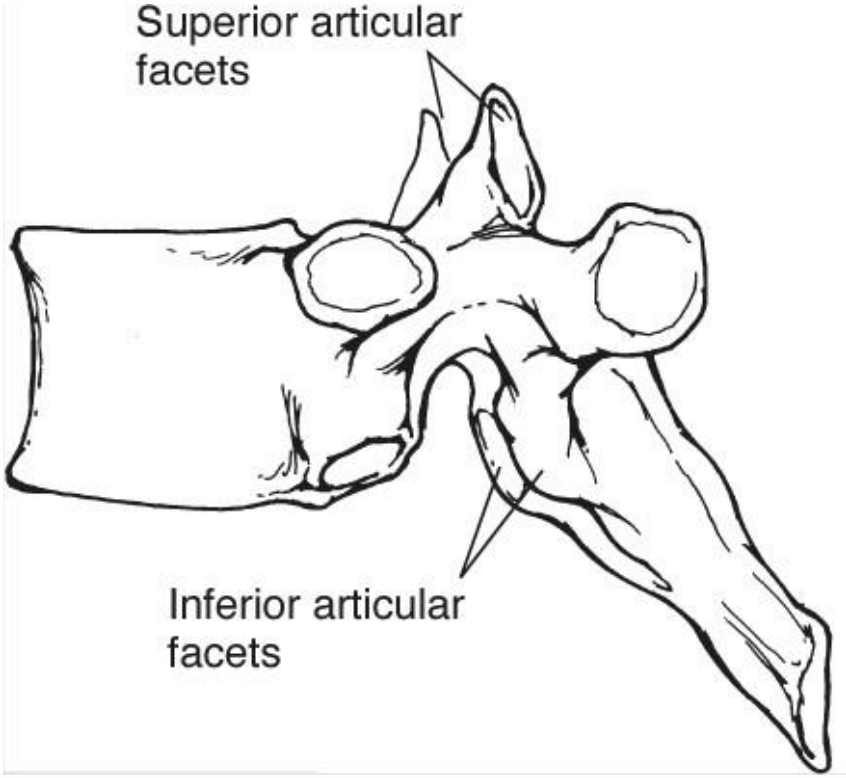


STRUCTURAL ORGANIZATION



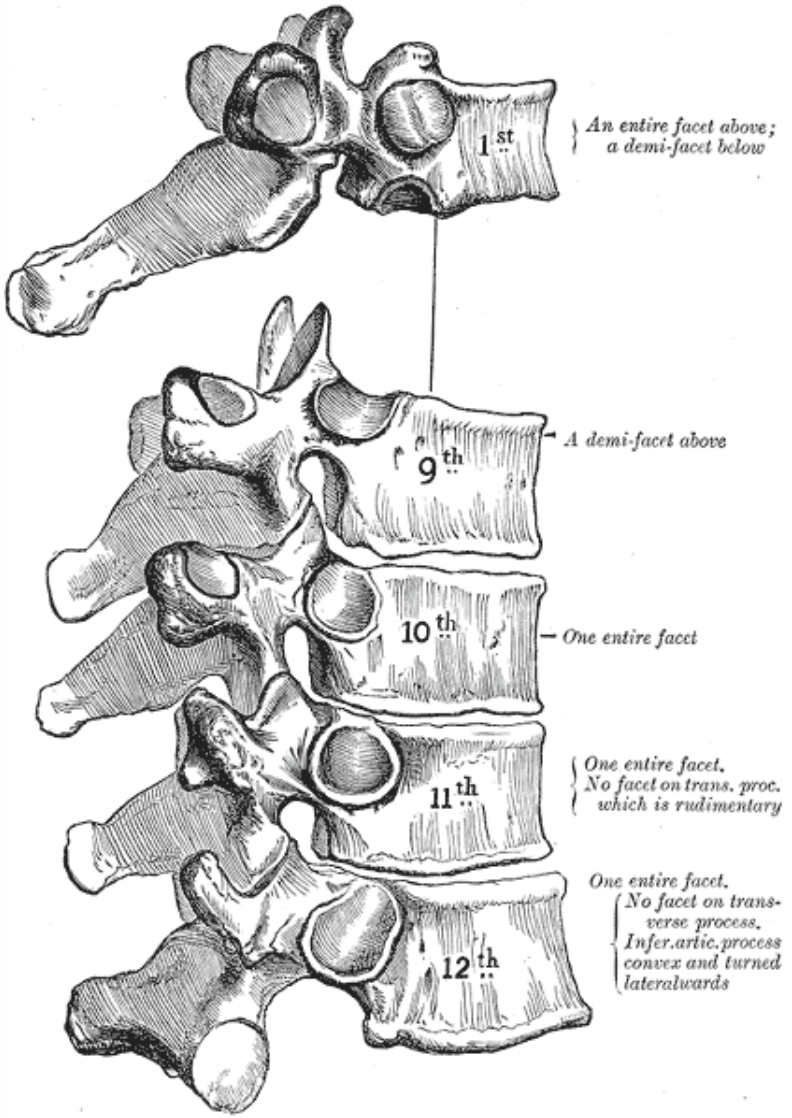
TYPICAL VERTEBRA

- T2-T8
  - Superior facets face posteriorly and are directed superolaterally
  - Inferior facets face anteriorly and are directed inferomedially
- Coronal plane orientation
  - 70-80º from transverse plane
  - 10-15° from frontal plane

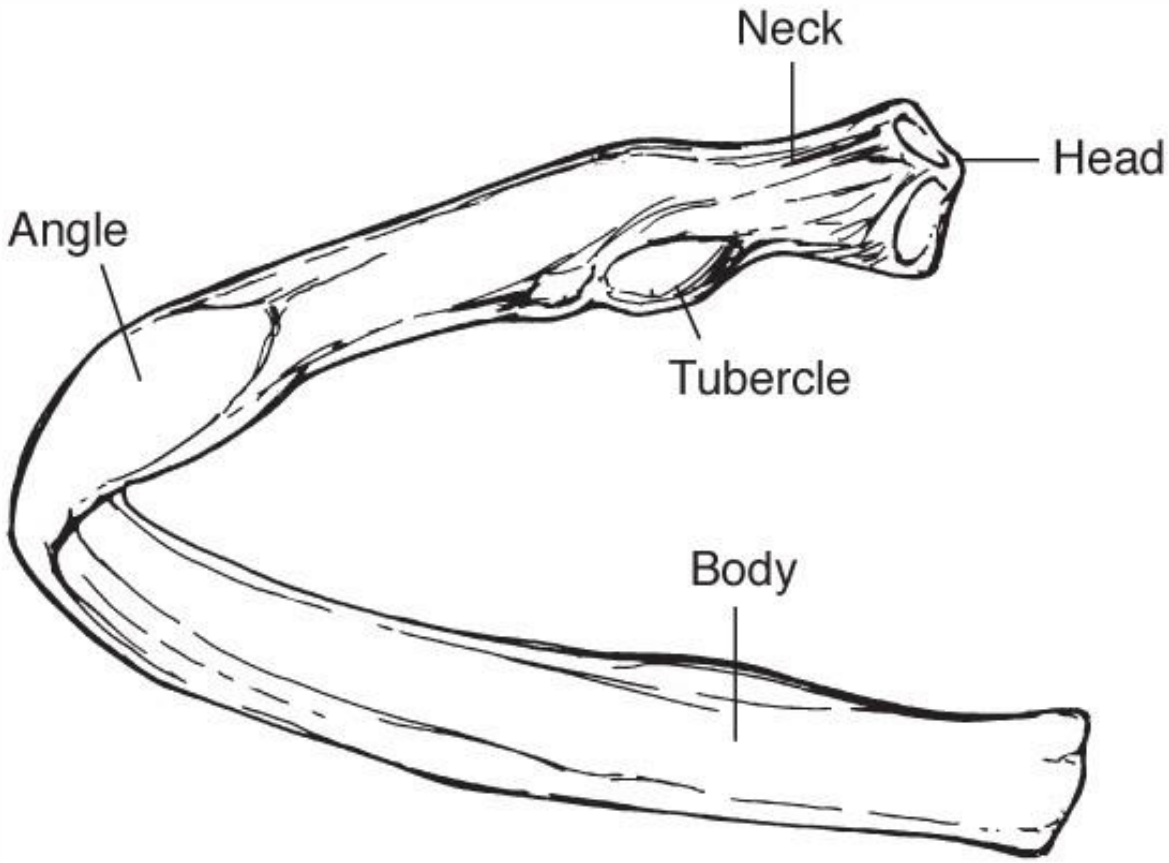


ATYPICAL THORACIC VERTEBRA

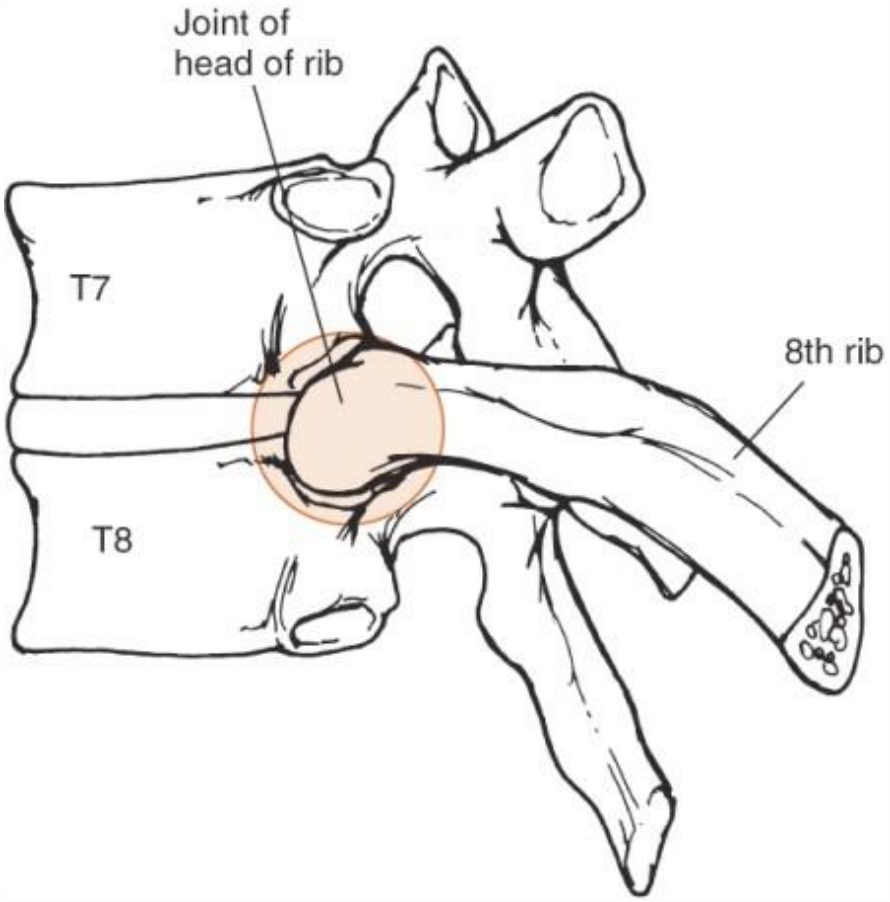
- One complete facet at the base of pedicle
  - Articulates with rib of same level
  - T11 and T12 lack costal facets on their transverse processes

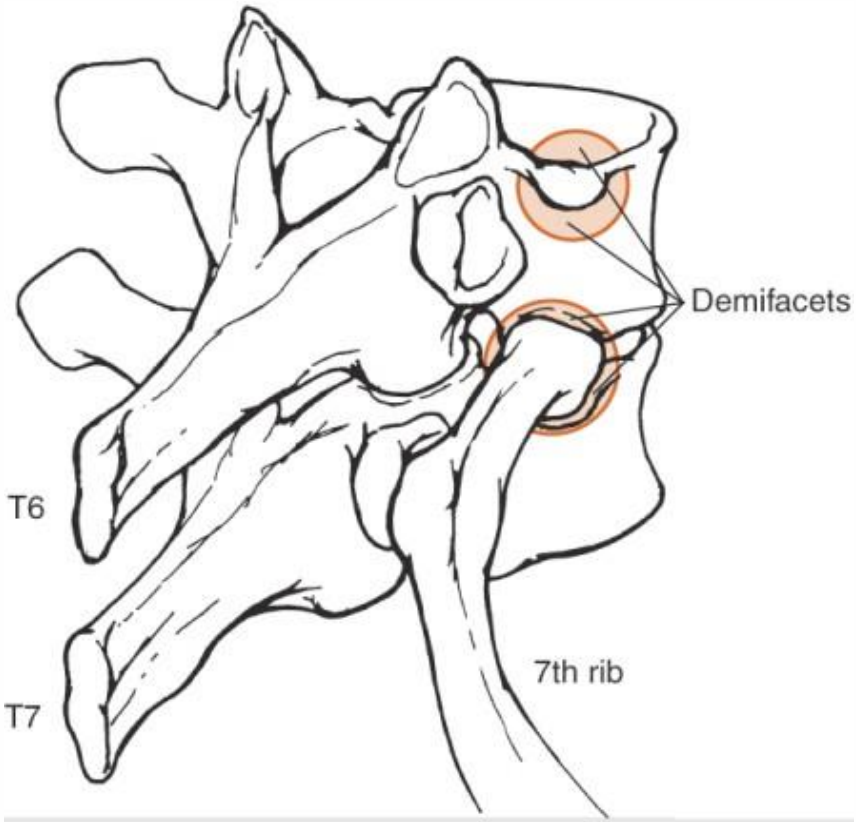


TYPICAL RIB

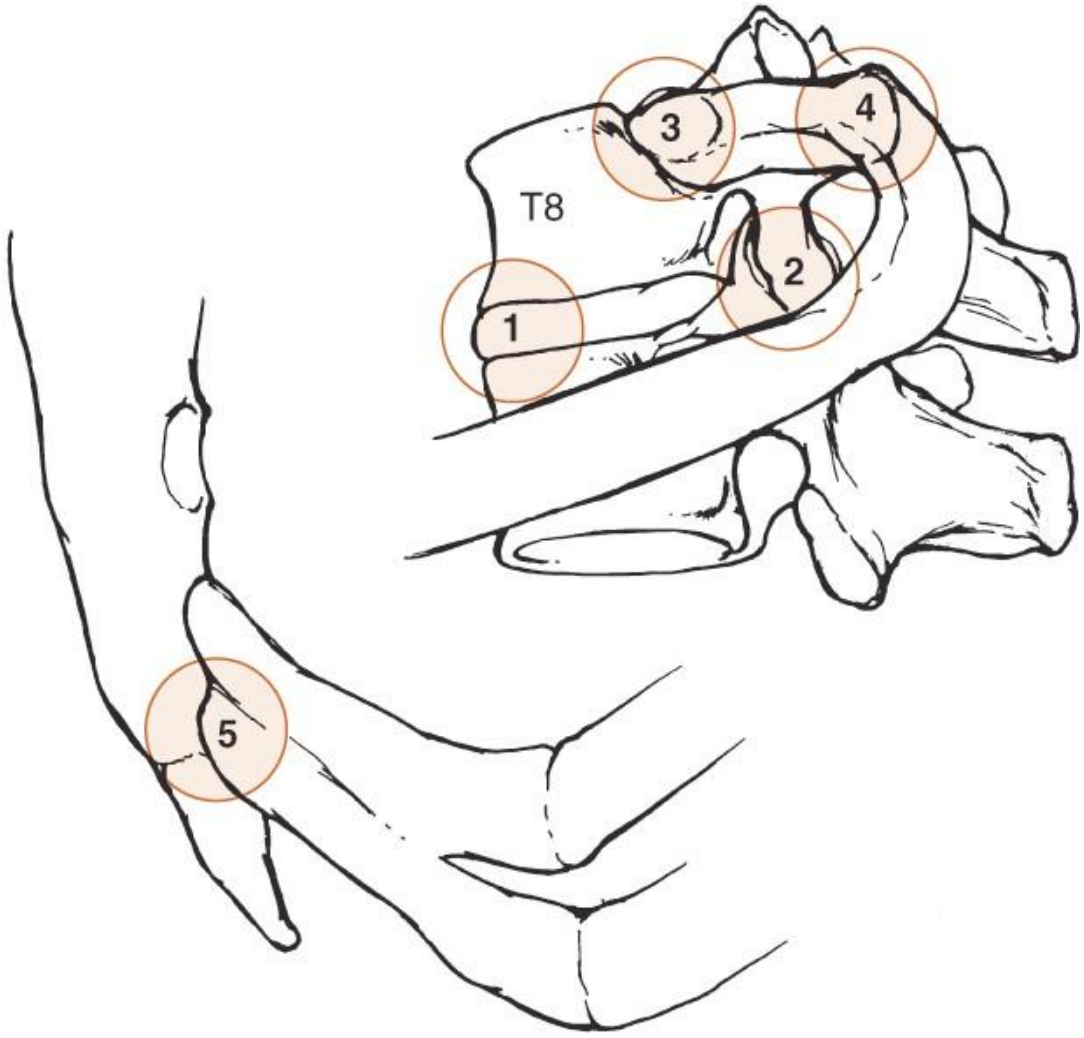


COSTOVERTEBRAL ARTICULATIONS

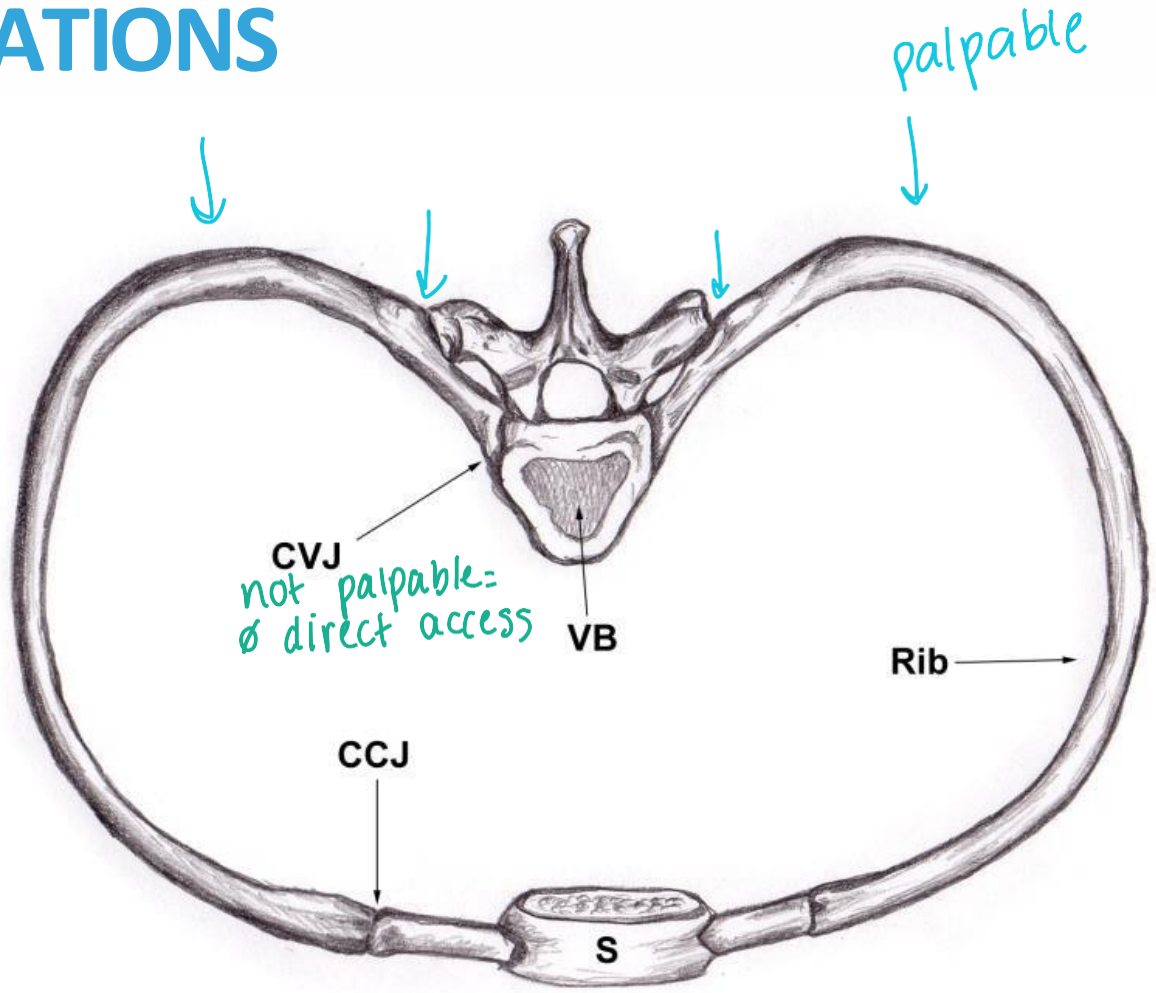




THORACIC ARTICULATIONS

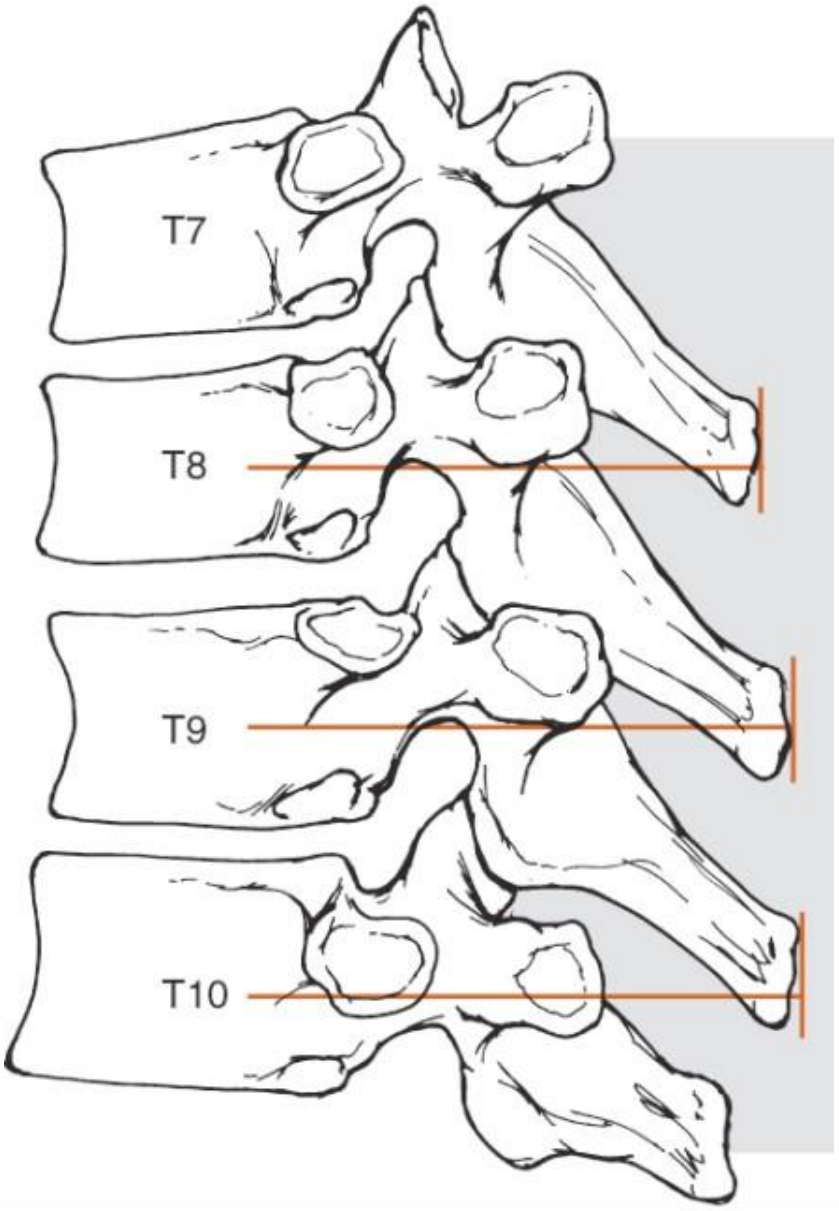


STRUCTURAL ORGANIZATION: ARTICULATIONS



RULE OF THREES

- **T1-T3 spinous processes**
  - SAME LEVEL as respective TPs
- **T4-T6 spinous processes**
  - ½ A VERTEBRAL LEVEL BELOW their respective TPs
- **T7-T9 spinous processes**
  - ONE FULL VERTEBRAL LEVEL BELOW their respective TPs
- **T10** is *ONE FULL LEVEL BELOW*
- **T11** is *½ LEVEL BELOW*
- **T12** is at the *SAME LEVEL*



GENERAL RANGE OF MOTION

- Flexion: 20-45 degrees
- Extension: 15-20 degrees
- Rotation: 45-60 degrees
- Side flexion: 25-45 degrees

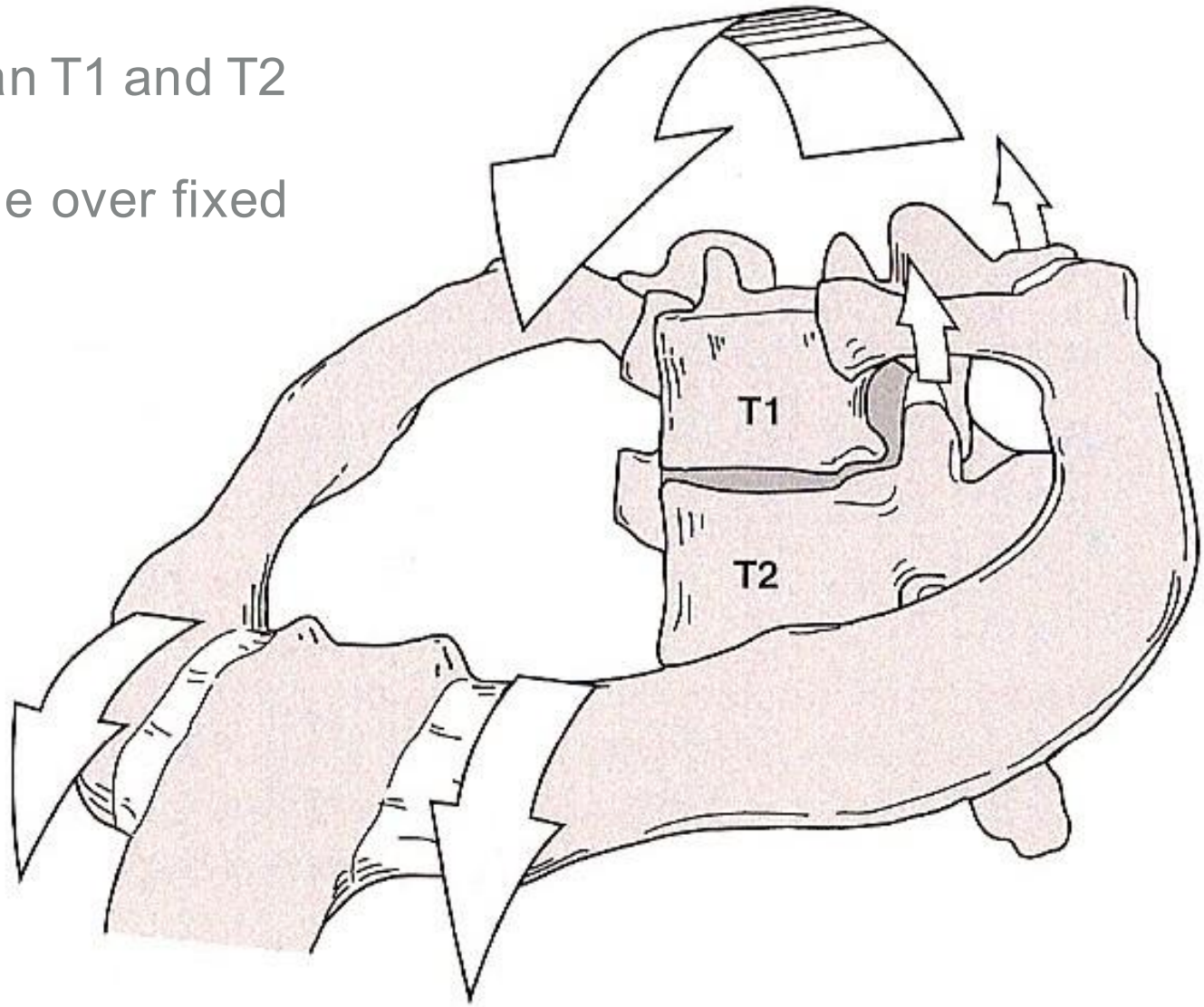


TALKING POINTS

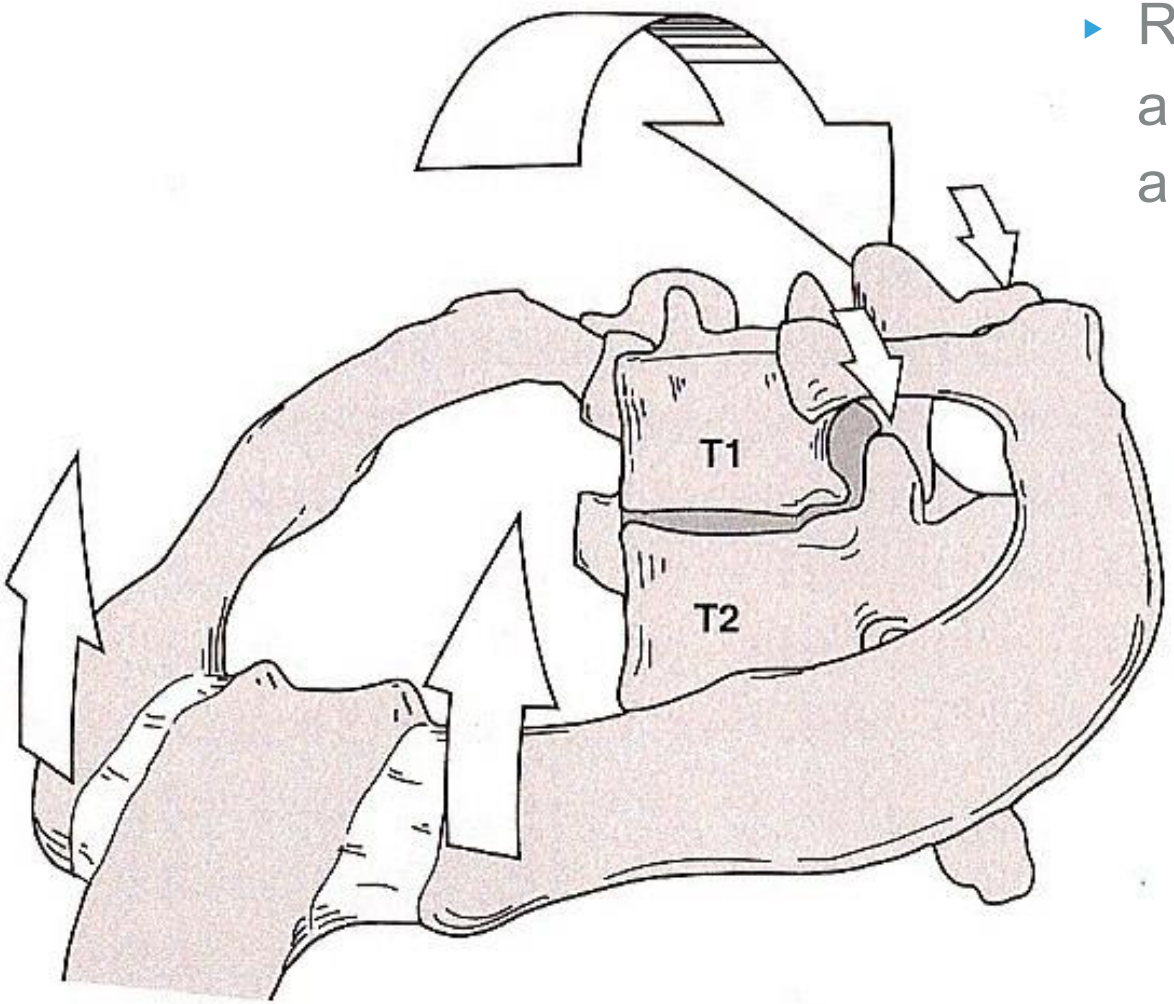
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VERTEBROMANUBRIAL FLEXION

- Ribs less mobile than T1 and T2
- Vertebral facets glide over fixed ribs at end of range

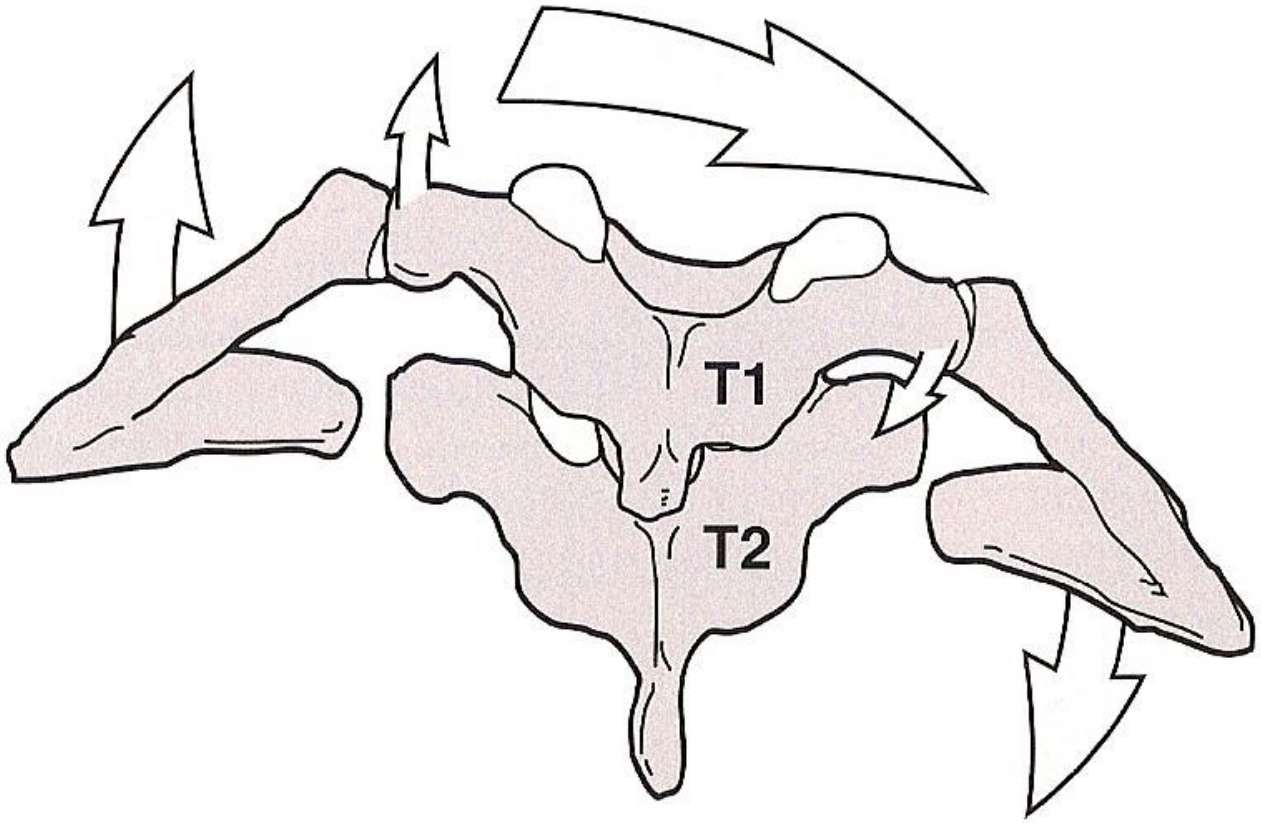


VERTEBROMANUBRIAL EXTENSION



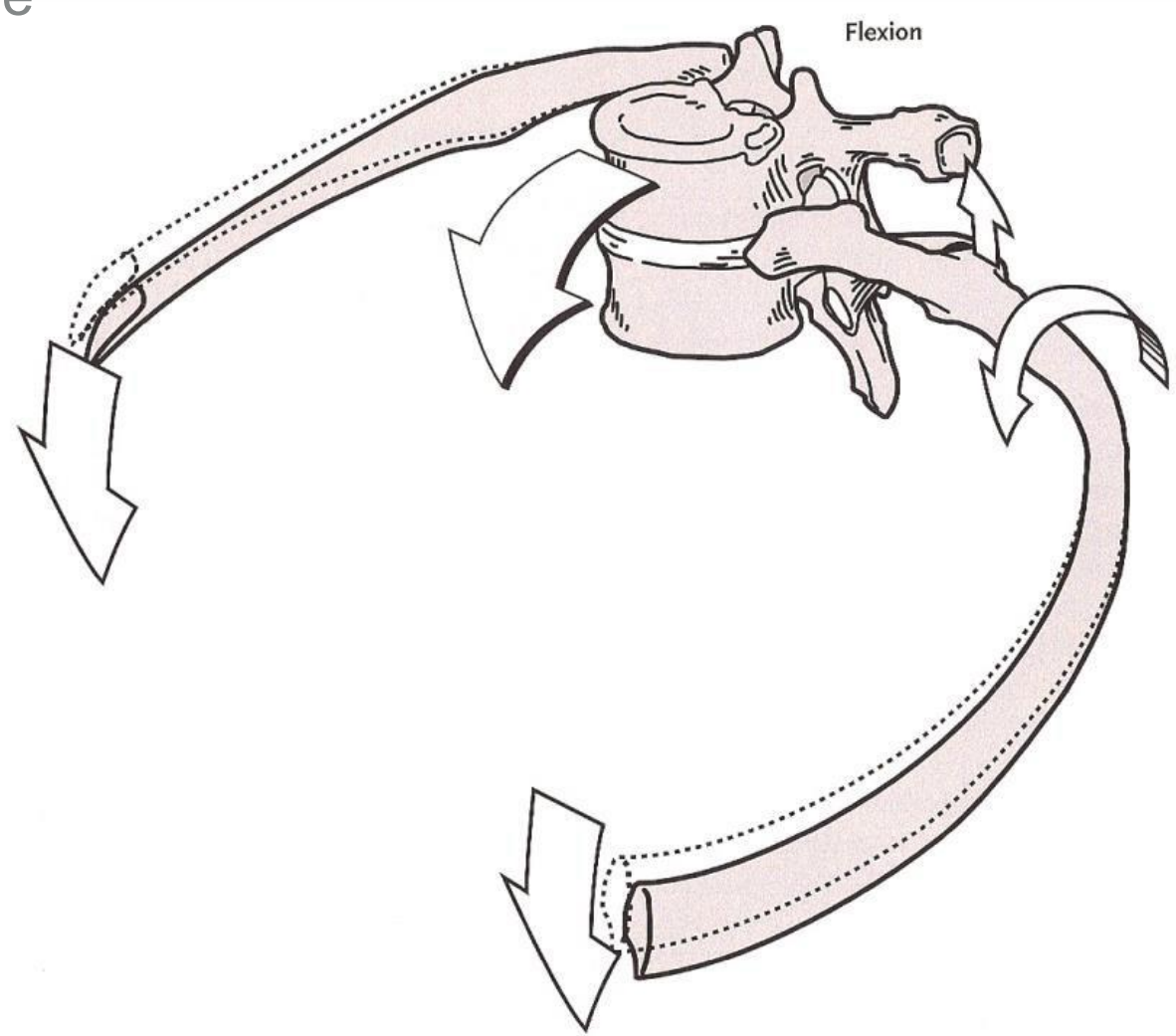
- Relative superior glide and anterior roll of the rib tubercle and the costotransverse joint

VERTEBROMANUBRIAL SIDE FLEXION AND ROTATION



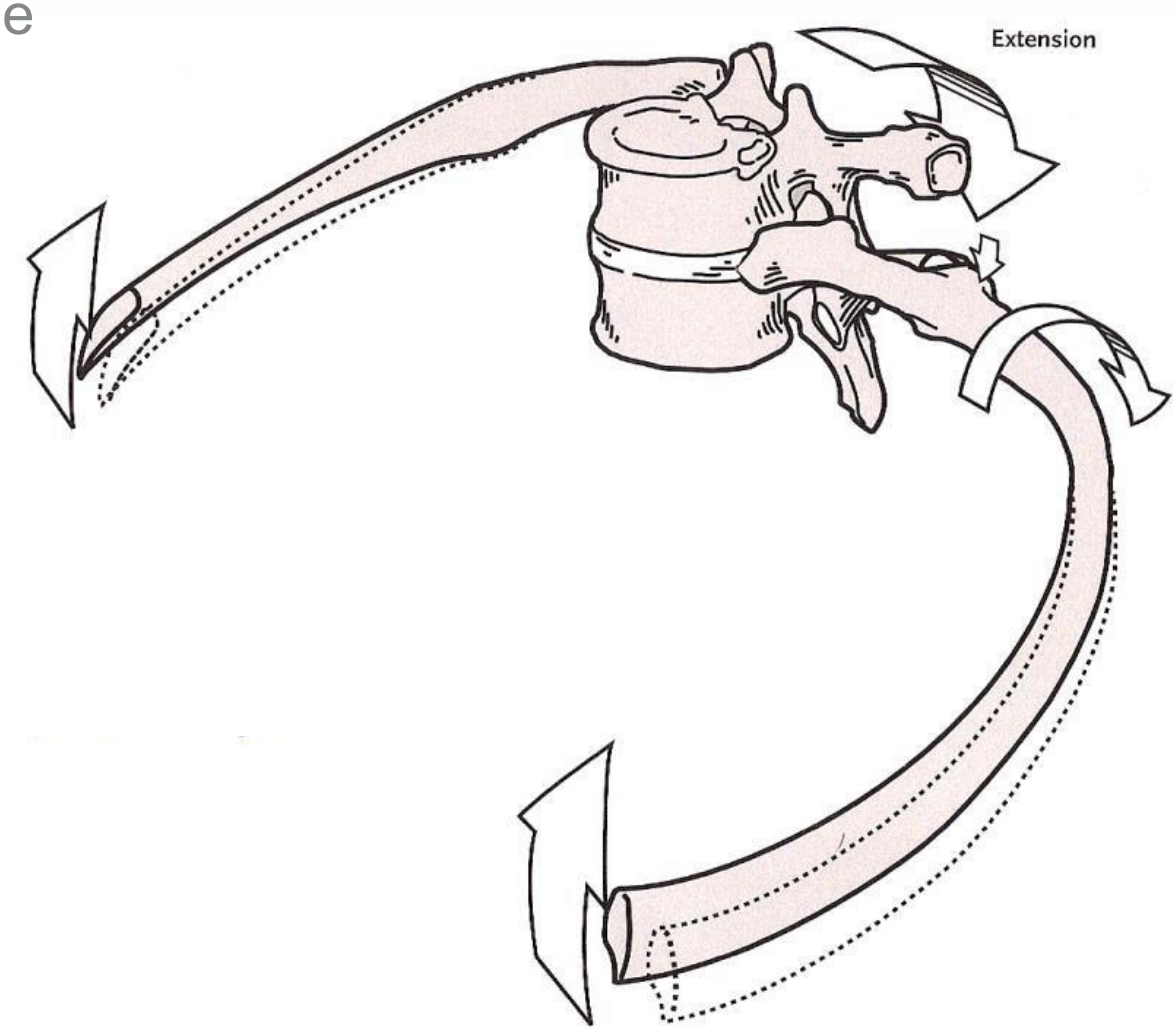
VERTEBROSTERNAL FLEXION

- Facet: Superoanterior glide
- Rib: Anterior rotation



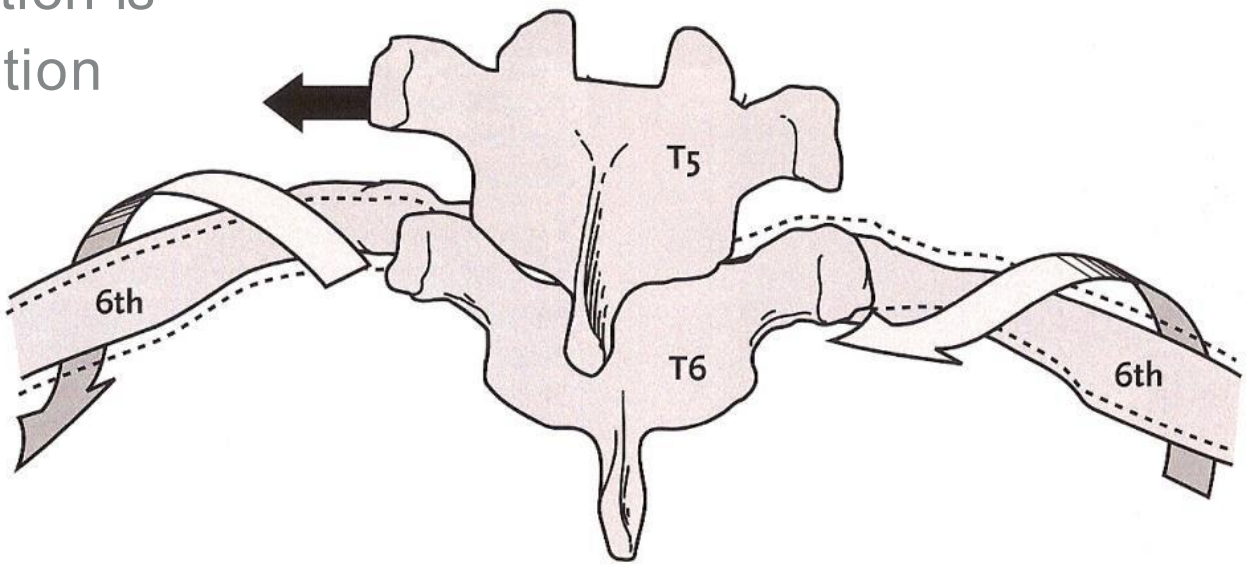
VERTEBROSTERNAL EXTENSION

- Facet: Inferoposterior glide
- Rib: Posterior rotation



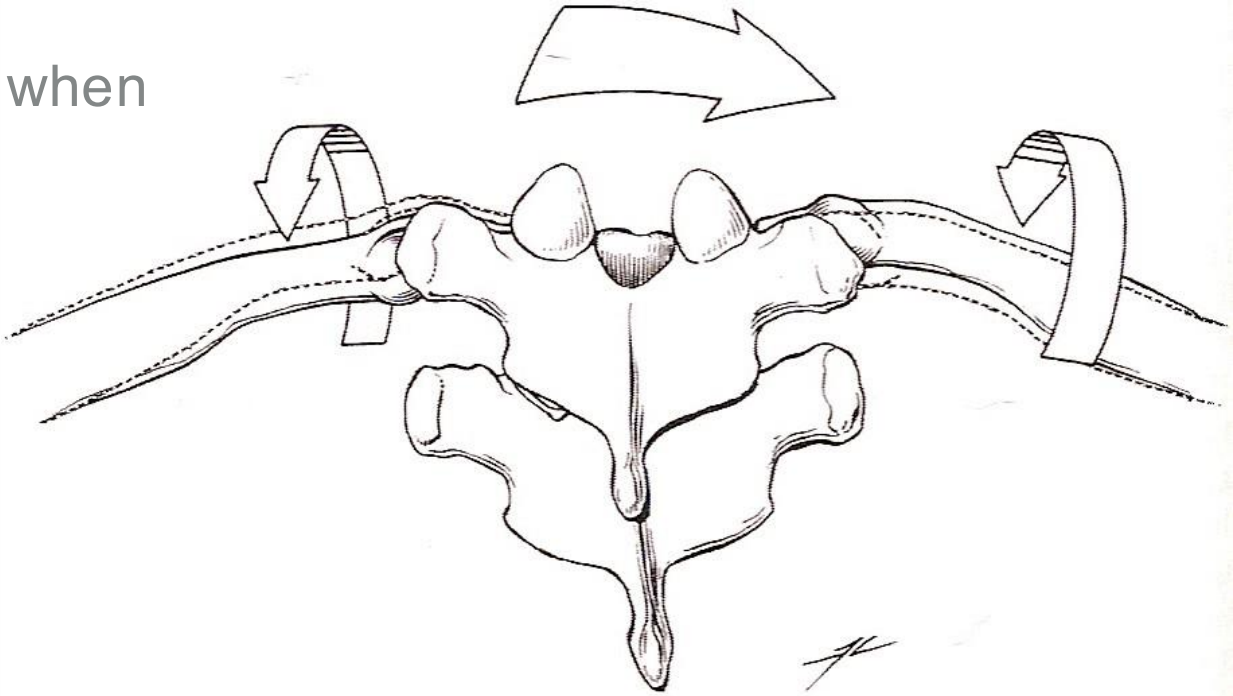
VERTEBROSTERNAL ROTATION

- Rotexion
  - Ipsilateral coupling of rotation and side flexion
  - Occurs when axial rotation is the first introduced motion



VERTEBROSTERNAL SIDEFLEXION

- Latexion
  - Contralateral coupling of side flexion and rotation
  - Occurs at end of range when side flexion is the first introduced motion

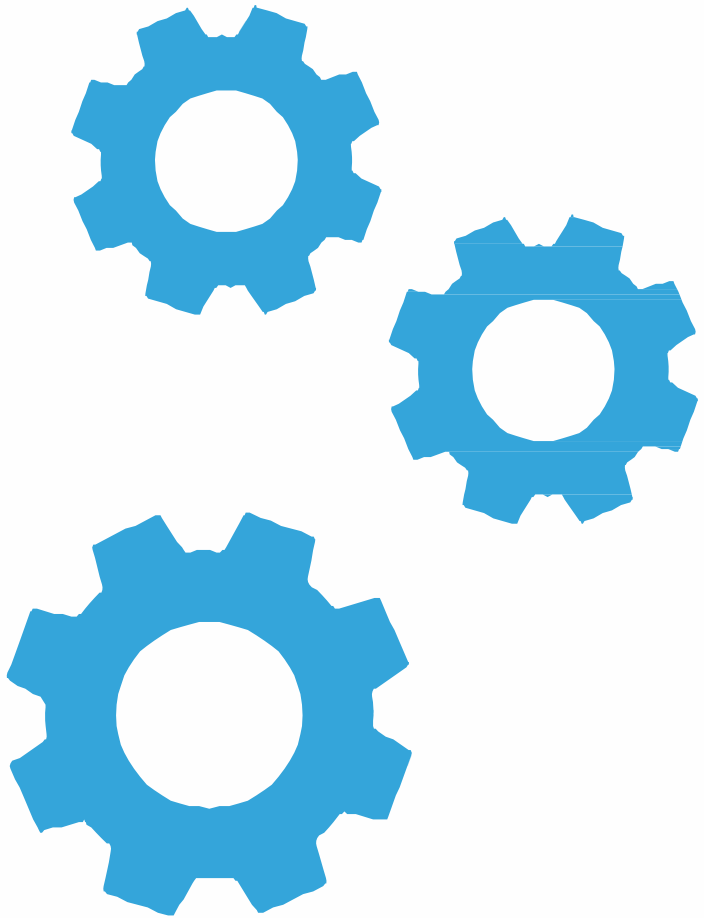


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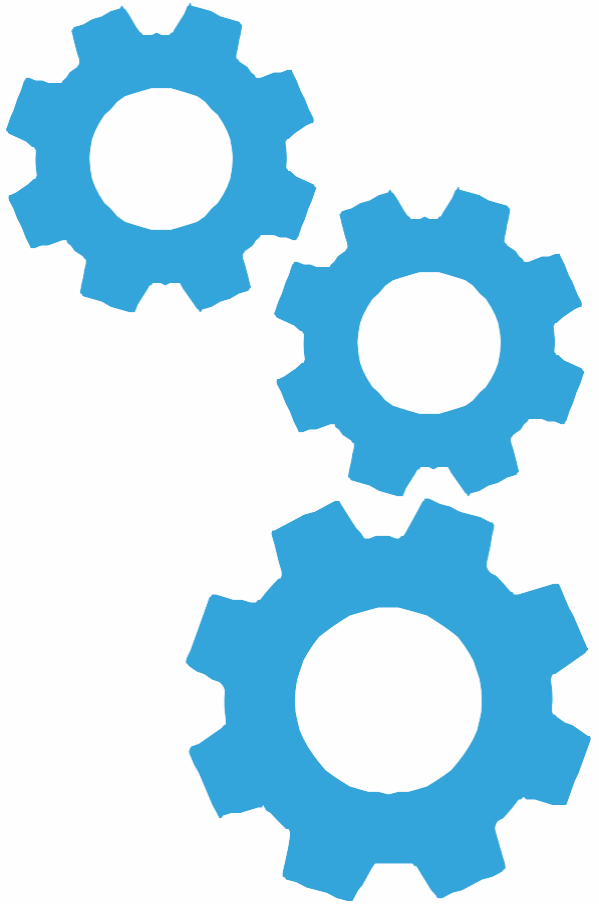
GENERAL VERTEBROCHONDRAL MOTION

- Designed to rotate with minimal restriction from costal elements
- Increased rib mobility
  - Costocartilages of ribs 8-10 are less firmly attached to sternum



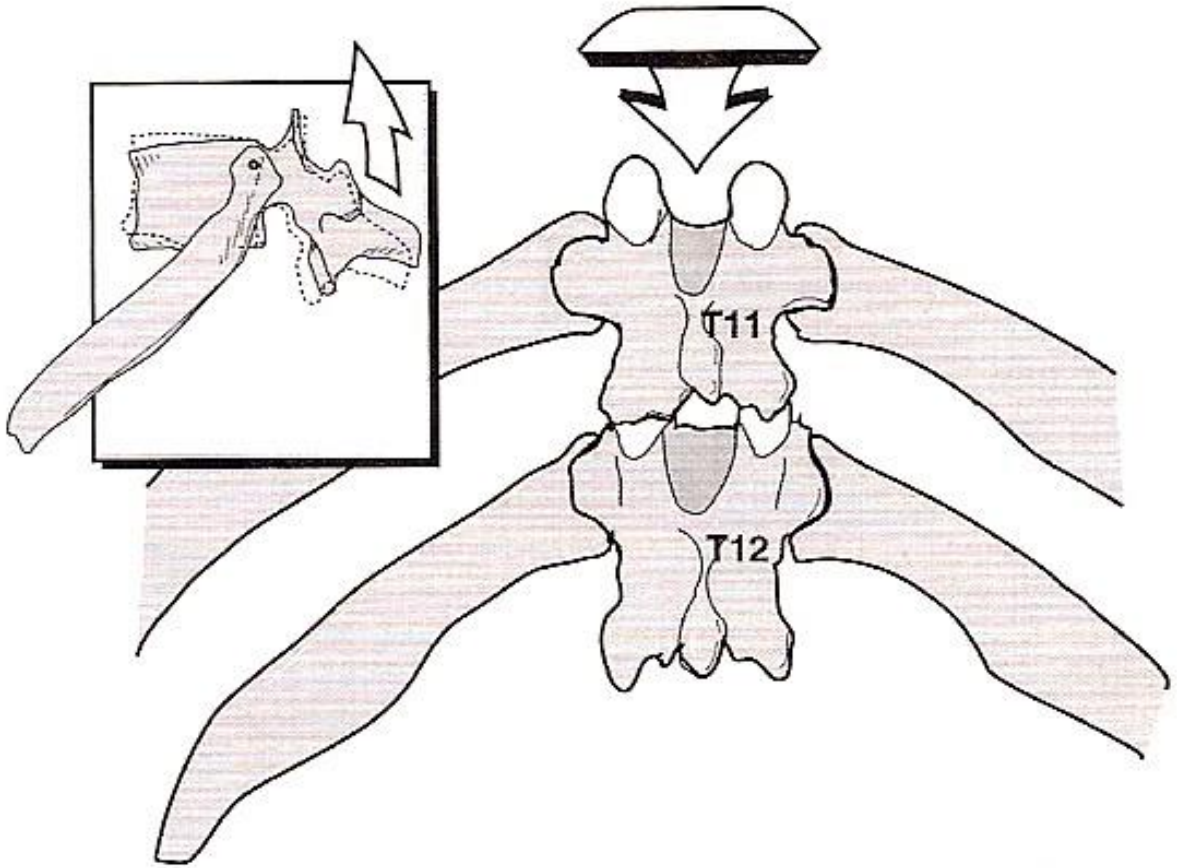
GENERAL THORACOLUMBAR MOTION

- Greatest flexibility in motion patterning
  - No costotransverse joint
  - No anterior rib attachment
- T10 and T11 are designed to rotate with minimal rib restriction



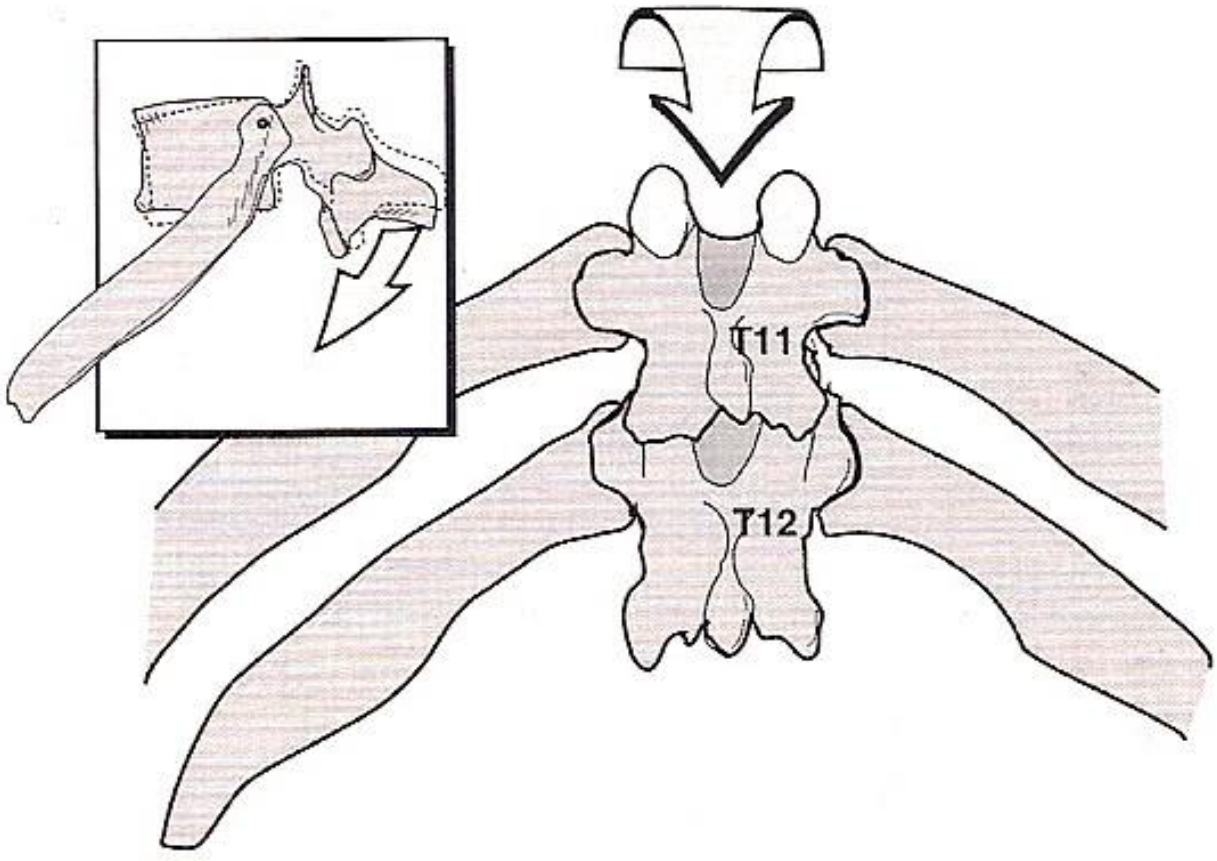
THORACOLUMBAR FLEXION

- Anterior sagittal rotation and anterior translation of superior vertebra
- Facets glide superiorly and anteriorly



THORACOLUMBAR EXTENSION

- Posterior sagittal rotation and posterior translation of superior vertebra
- Facets glide inferiorly and posteriorly

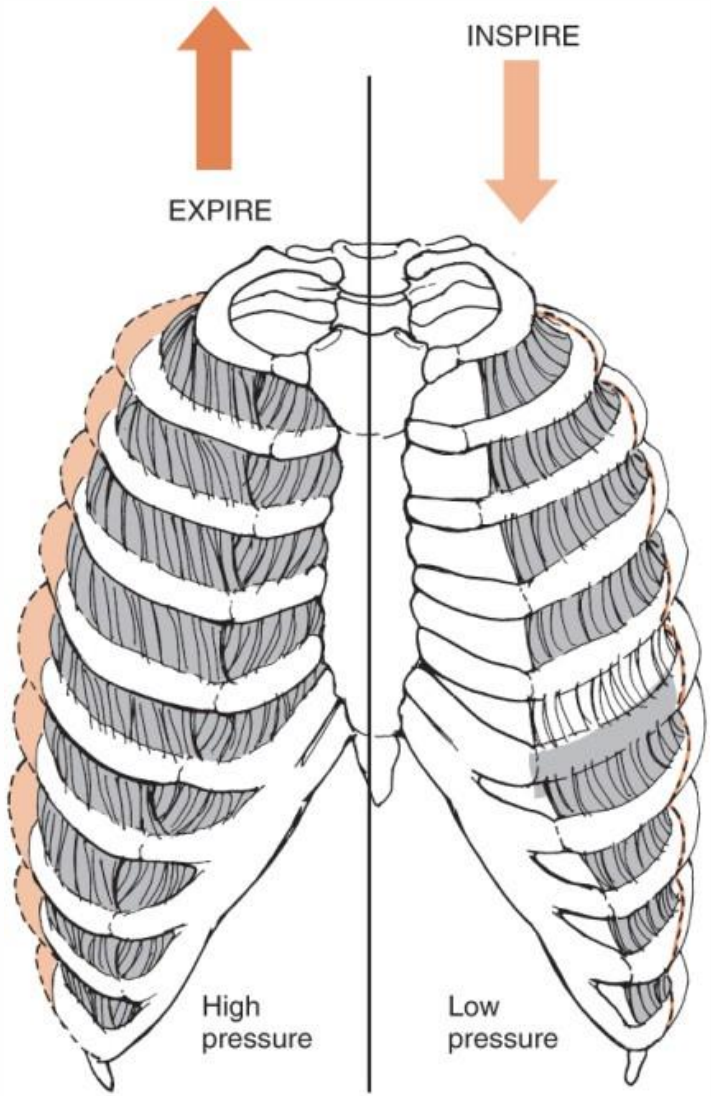


THORACOLUMBAR ROTATION & SIDEFLEXION

- Pure axial rotation can occur at the T11/12 segment
  - The coronally oriented facets do not dictate specific coupling of sideflexion when rotation is induced

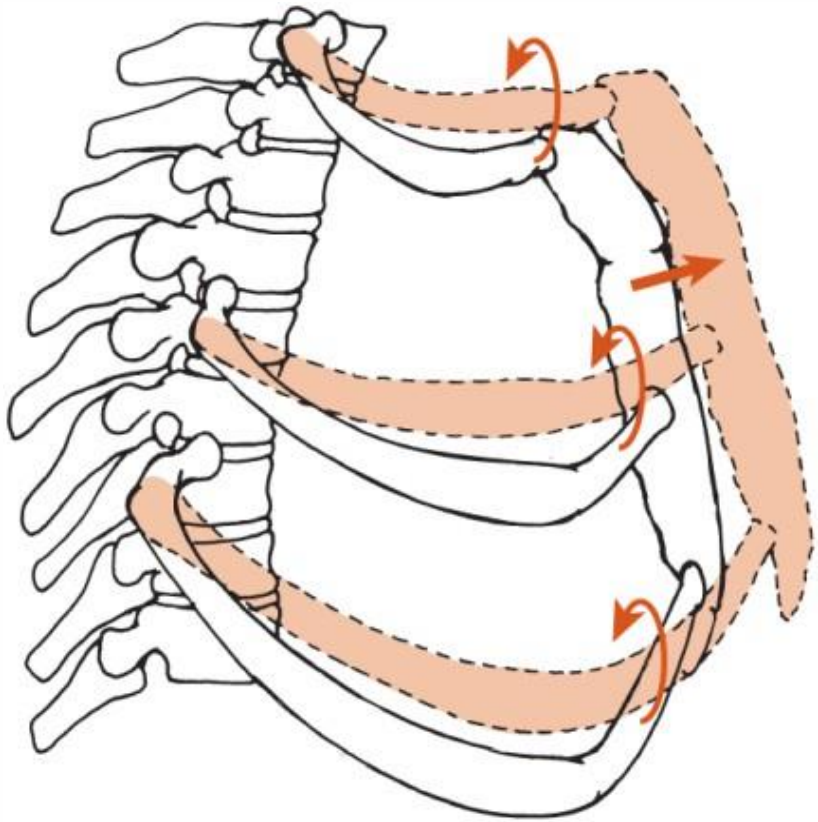
GENERAL MECHANICS OF RESPIRATION

- Intrathoracic pressure changes inversely with volume of thoracic cage
  - **Inspiration**
    - Diaphragm descends
    - Rib cage expands
    - V↑ → P↓
    - Air rushes in to fill the lungs
  - **Expiration**
    - Diaphragm relaxes
    - Rib cage contracts
    - V↓ → P↑
    - Air passively forced out of lungs

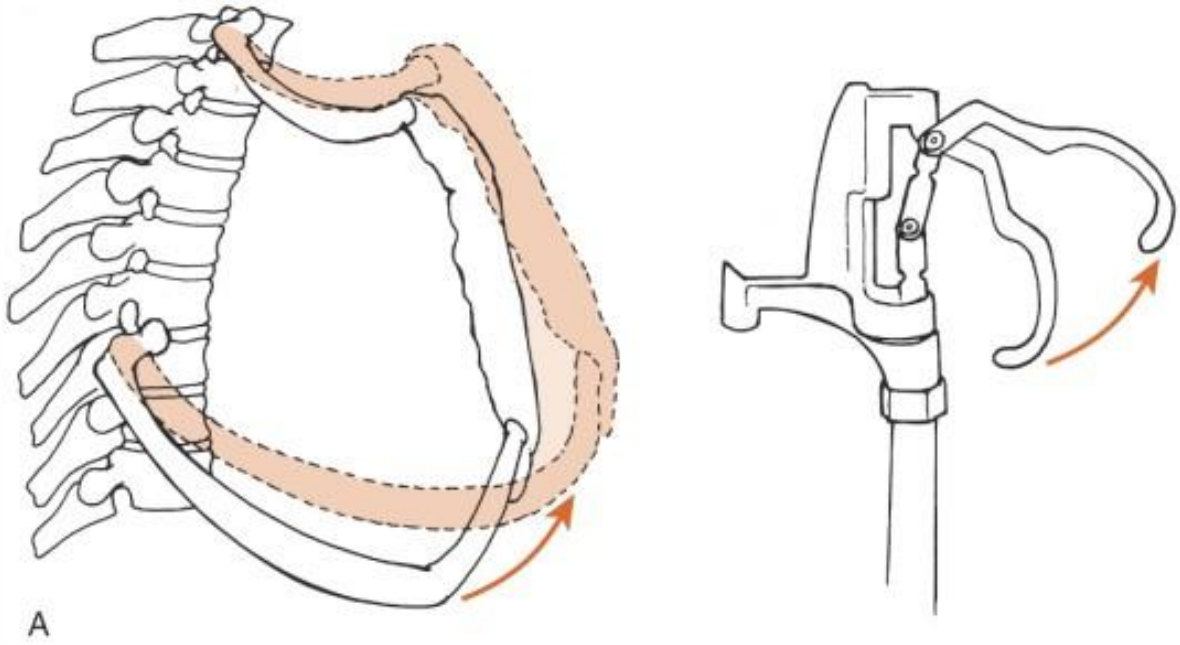


COSTAL CAGE MOTION DURING RESPIRATION

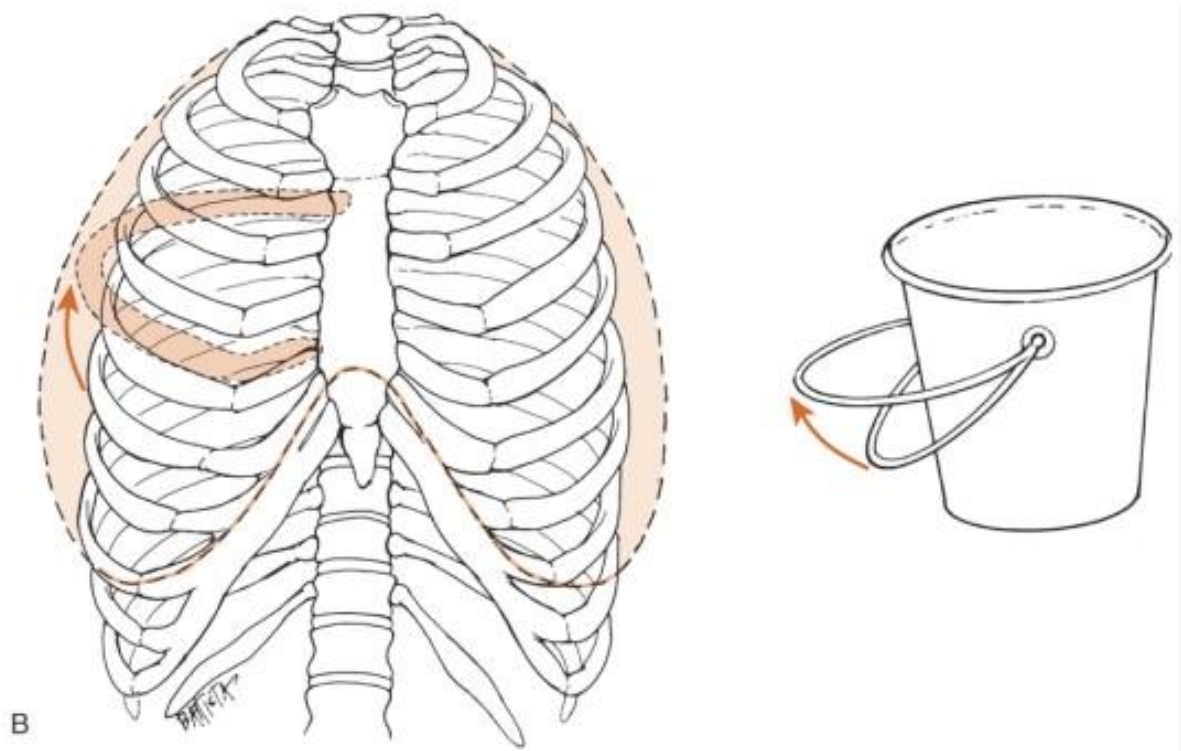
- Diaphragm contraction creates the following:
  - Elevates and posteriorly rotates the lower six ribs
- Rib elevation creates the following:
  - Anterosuperior movement of sternum
  - Torsion of costalcartilages



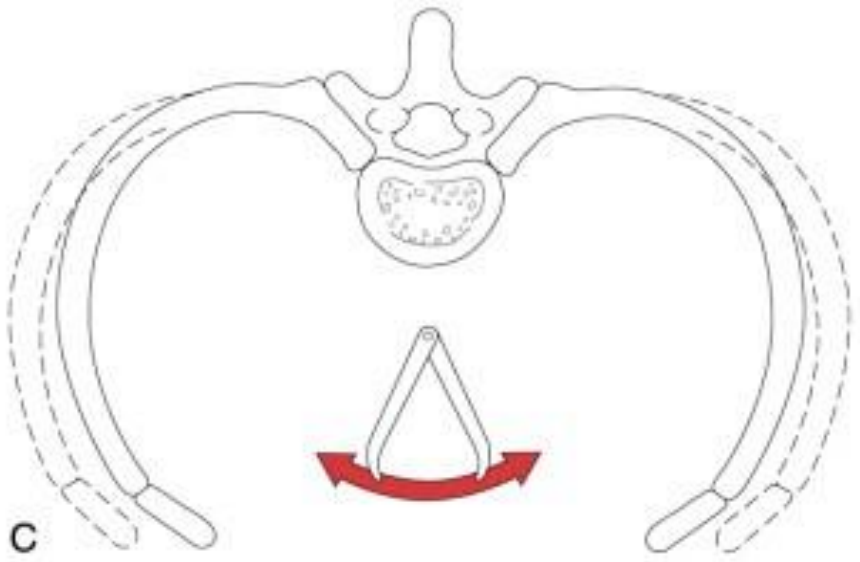
COSTAL CAGE MOTION DURING RESPIRATION



COSTAL CAGE MOTION DURING RESPIRATION



COSTAL CAGE MOTION DURING RESPIRATION



OSTEOKINEMATIC RIB MOTION

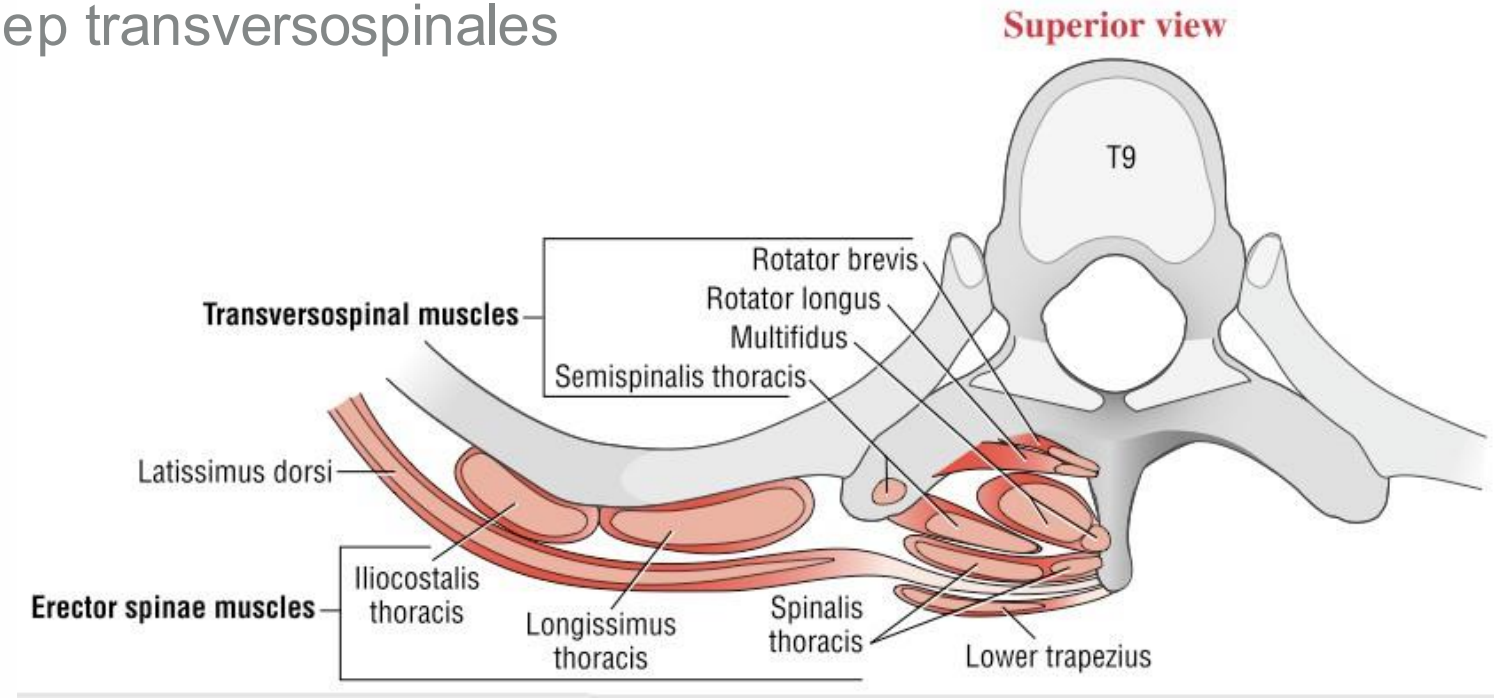
- Thoracic flexion
  - Anterior rib rotation
- Thoracic extension
  - Posterior rib rotation
- Thoracic sideflexion
  - Ipsilateral anterior rotation
  - Contralateral posterior rotation
- Thoracic rotation
  - Ipsilateral posterior rotation
  - Contralateral anterior rotation

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POSTERIOR THORACIC MUSCULATURE

- Superficial paraspinals
- Deep transversospinales



LOCAL STABILIZERS: THORAX

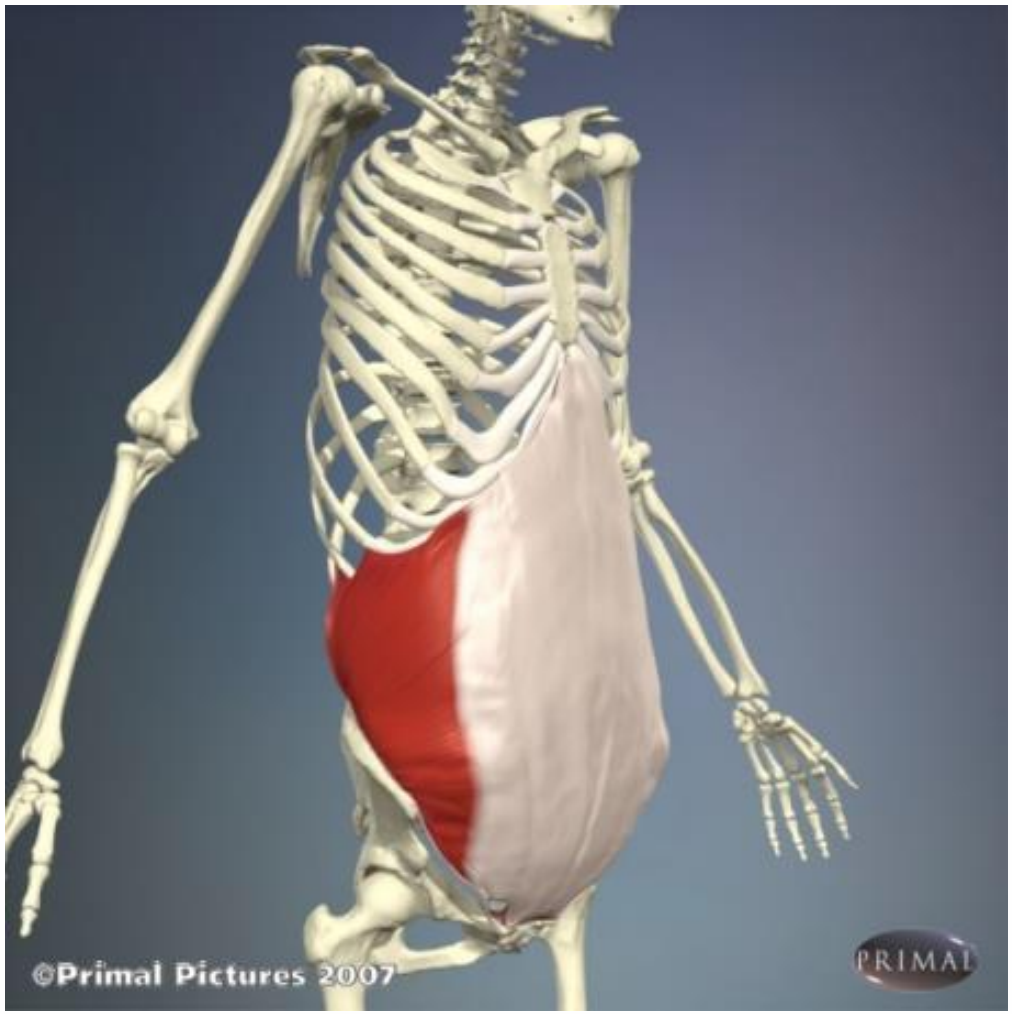


GLOBAL STABILIZERS: POSTERIOR THORAX



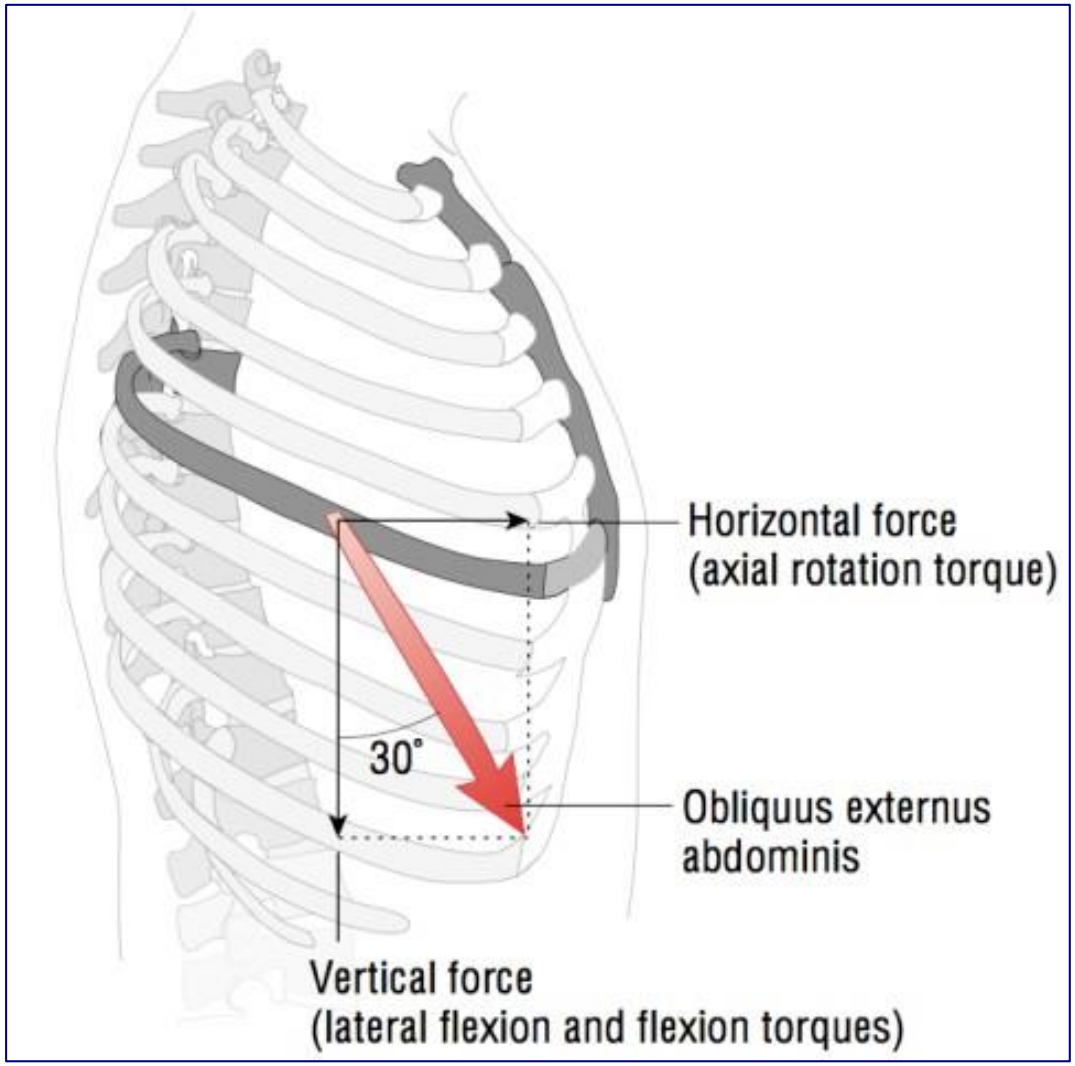
- Flexion control
  - Semispinalis thoracis
  - Spinalis thoracis
  - Superficial multifidus
- Rotation control
  - Semispinalis thoracis
  - Spinalis thoracis

GLOBAL STABILIZERS: ANTERIOR TRUNK



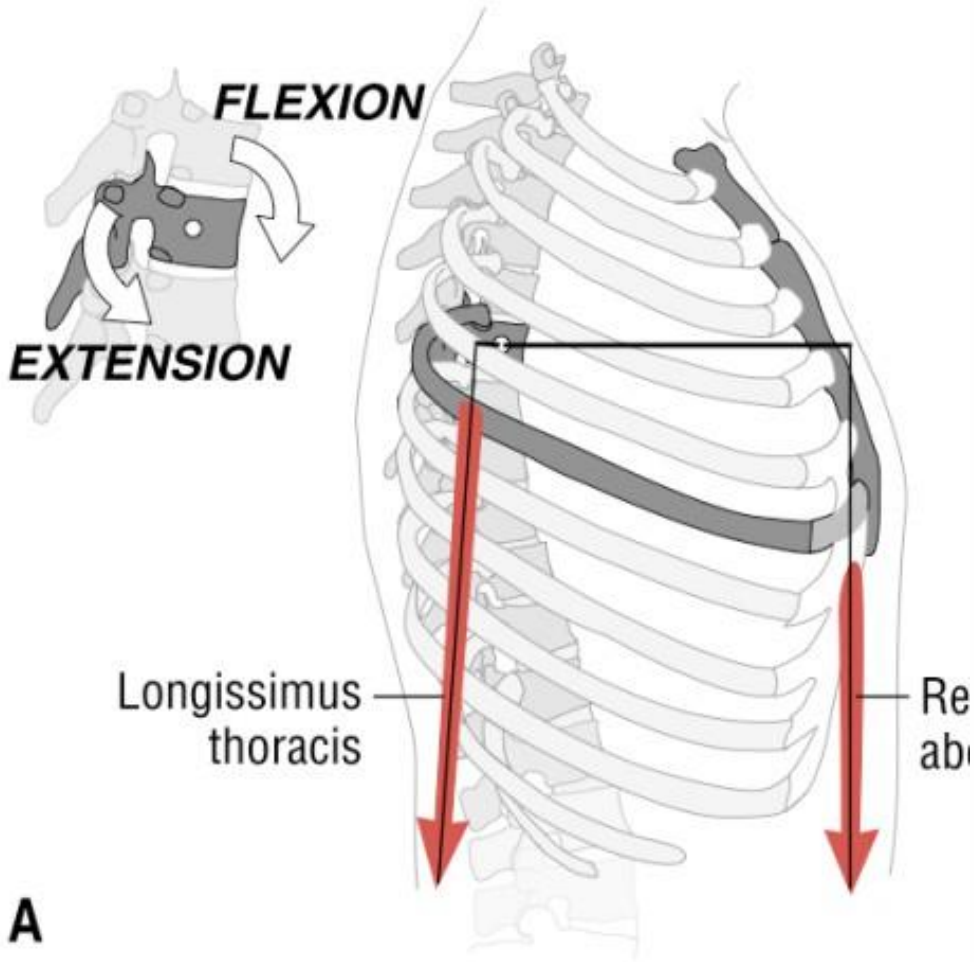


EXTERNAL OBLIQUE



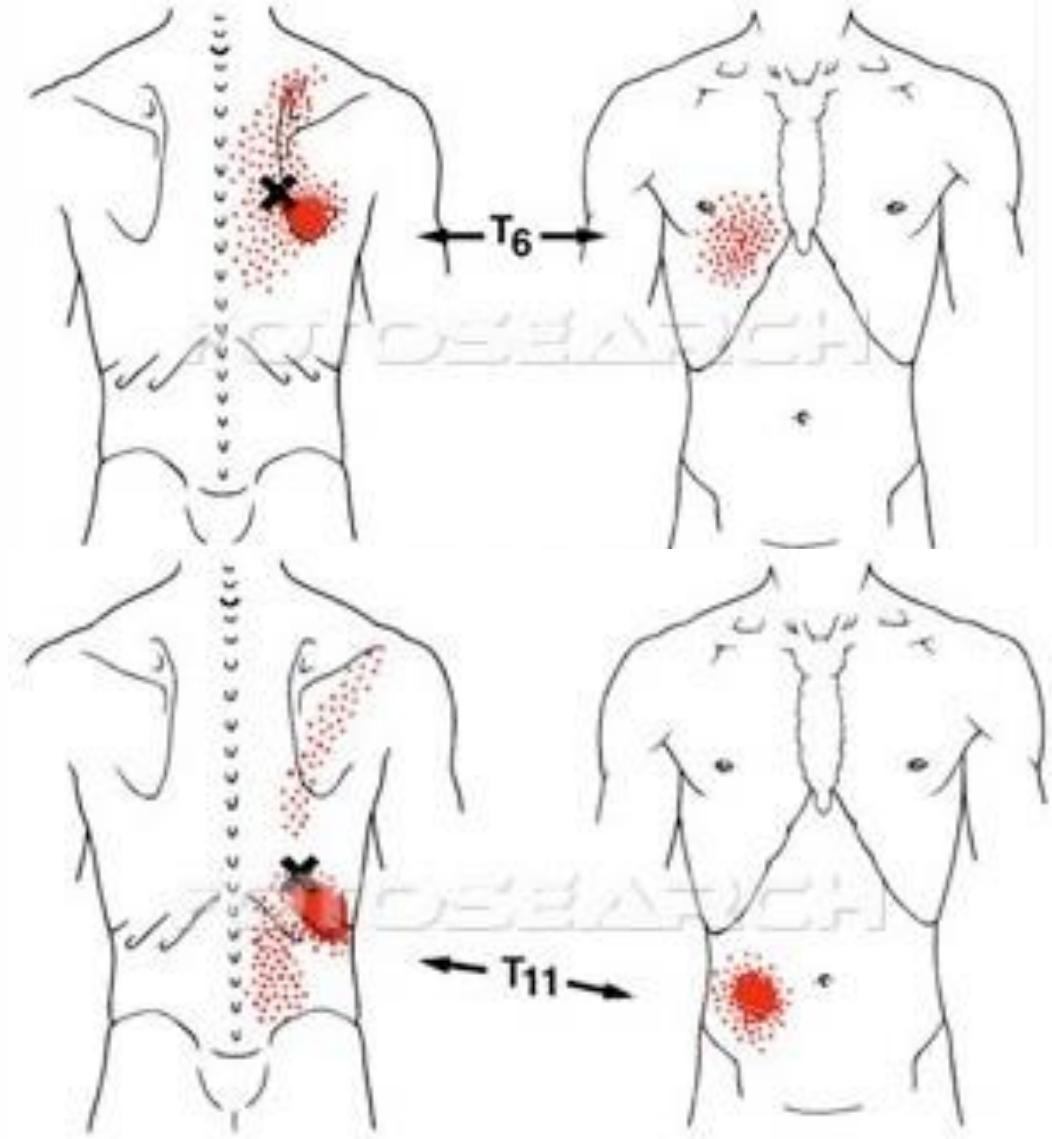
GLOBAL MOBILIZERS: THORAX & TRUNK

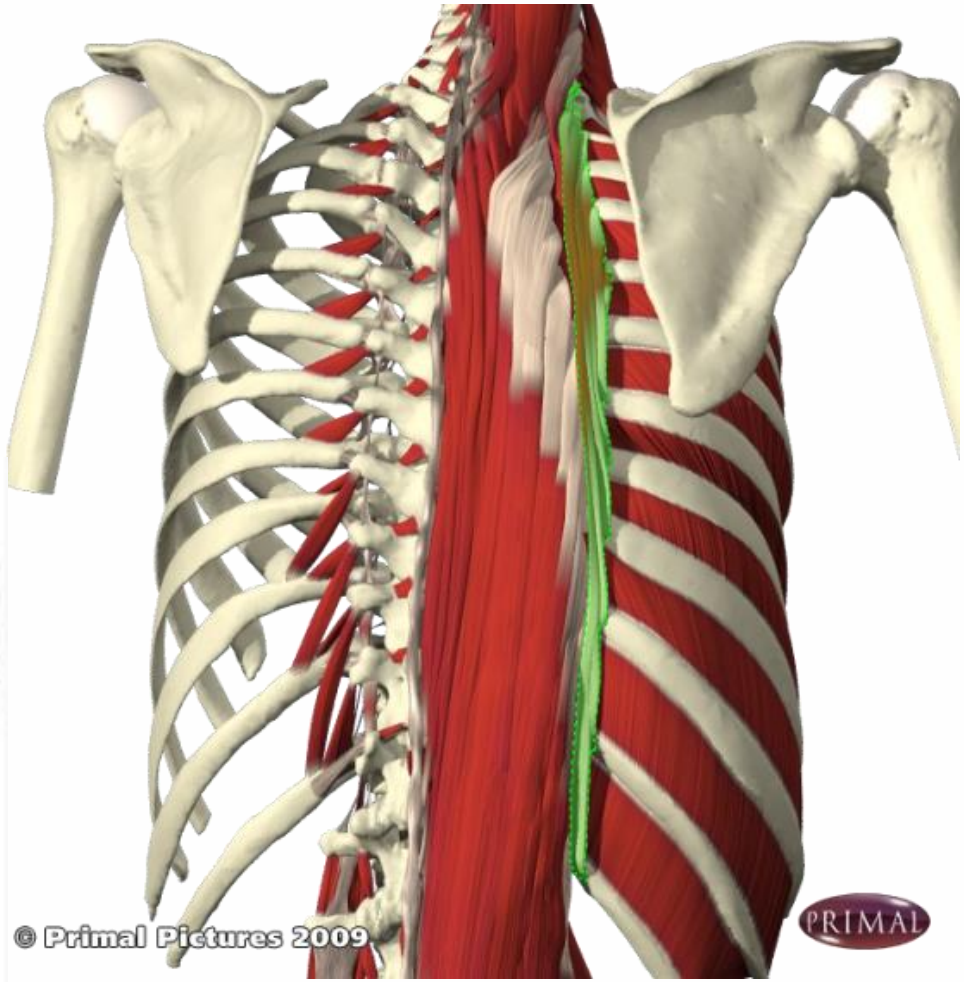
- Iliocostalis thoracis
- Iliocostalis lumborum
- Longissimus thoracis
- Rectus abdominis
- Quadratus lumborum
- Latissimus dorsi



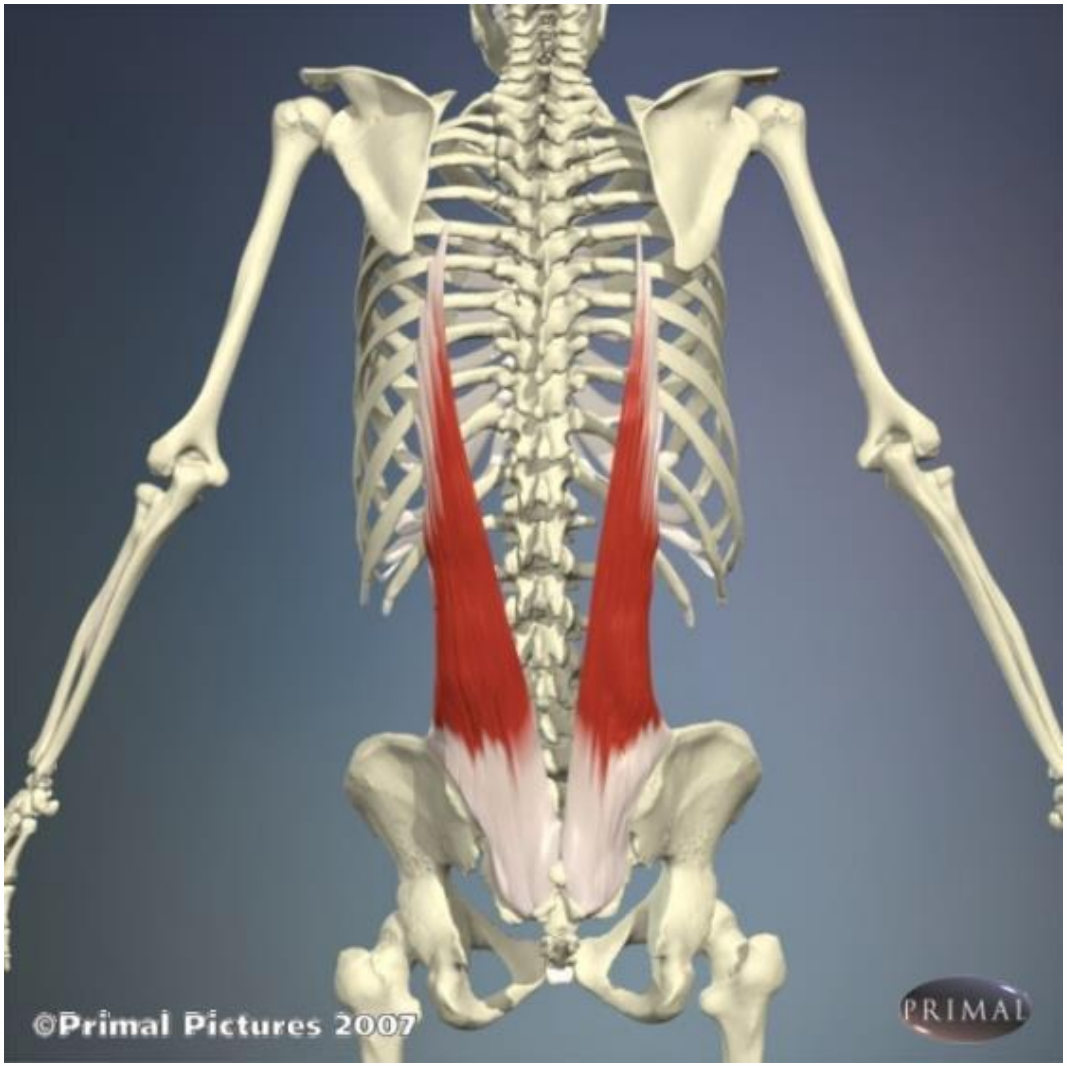
Sagittal Plane

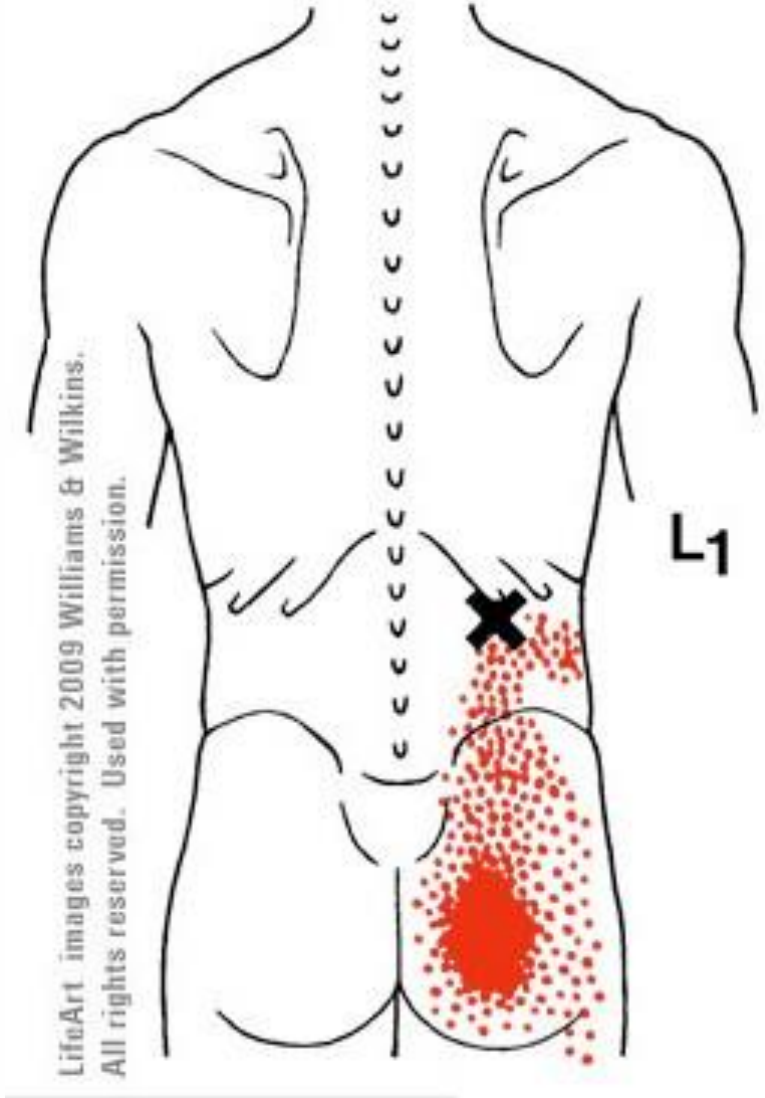
ILIOCOSTALIS THORACIS



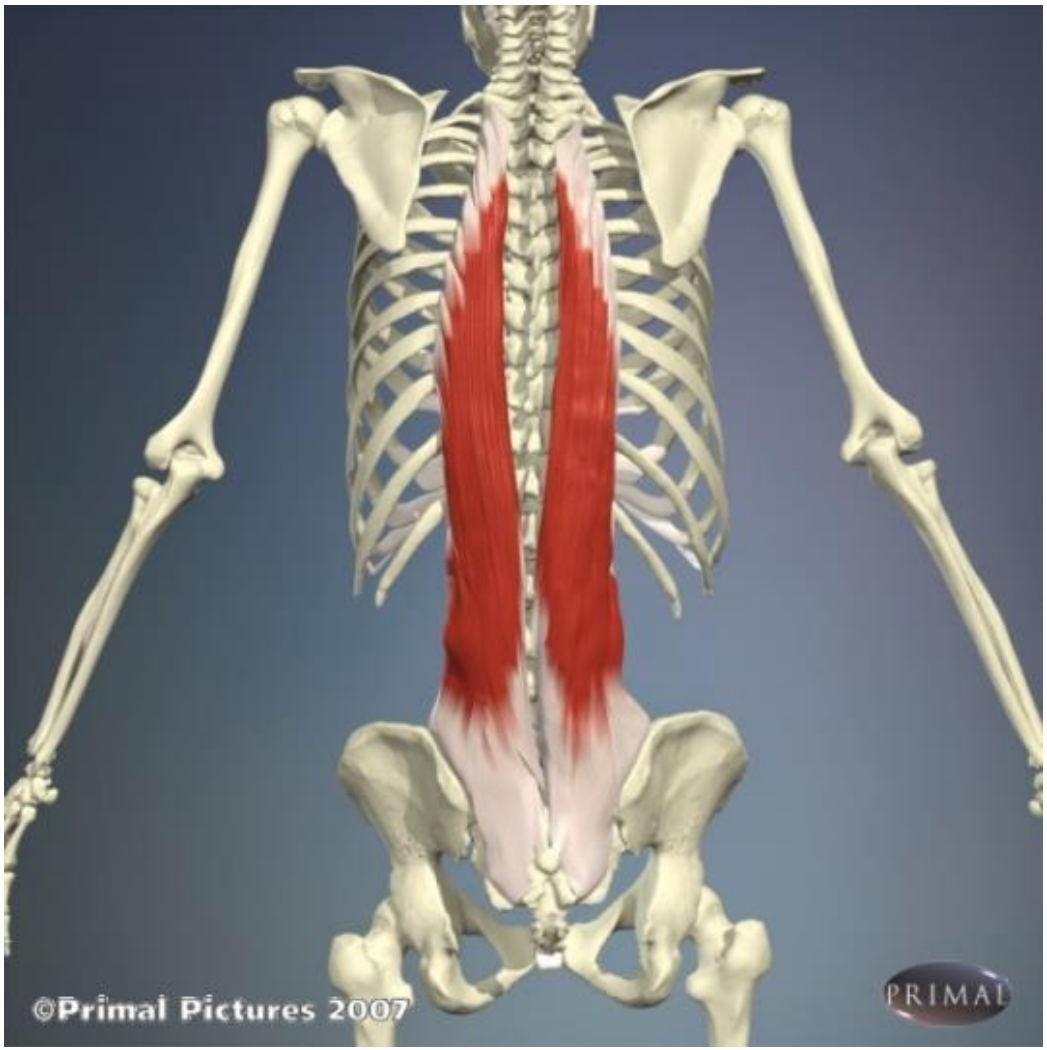


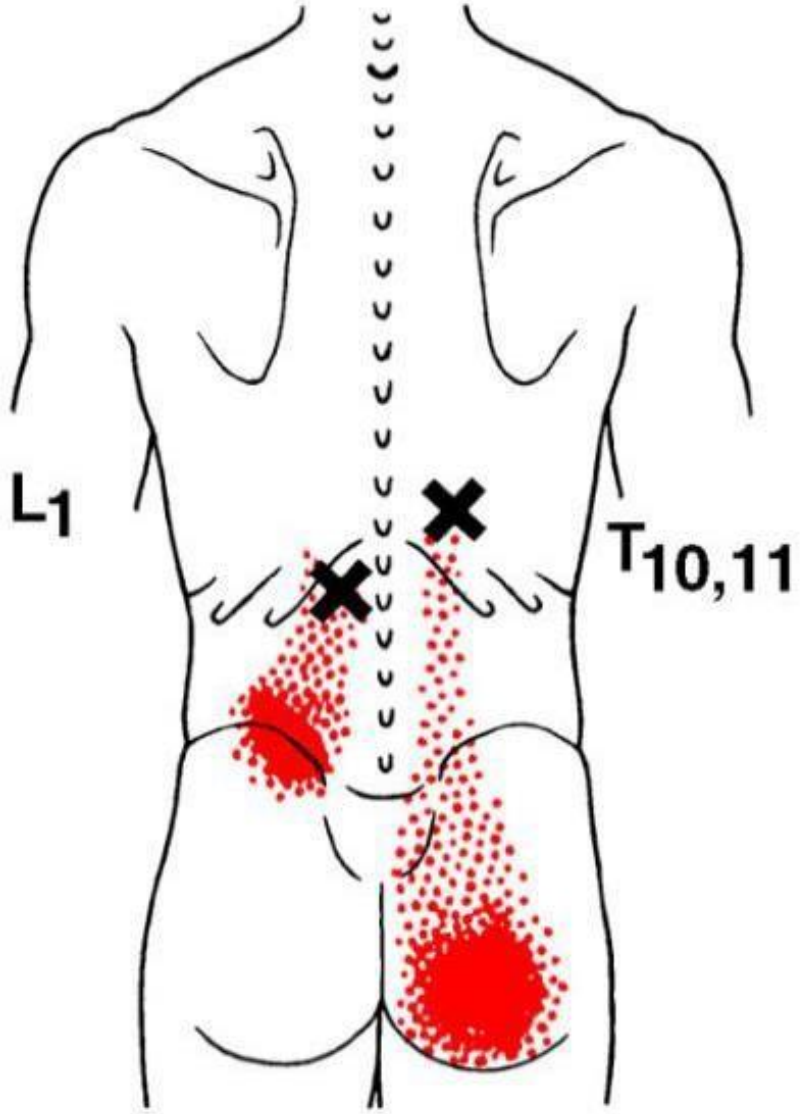
ILIOCOSTALIS LUMBORUM





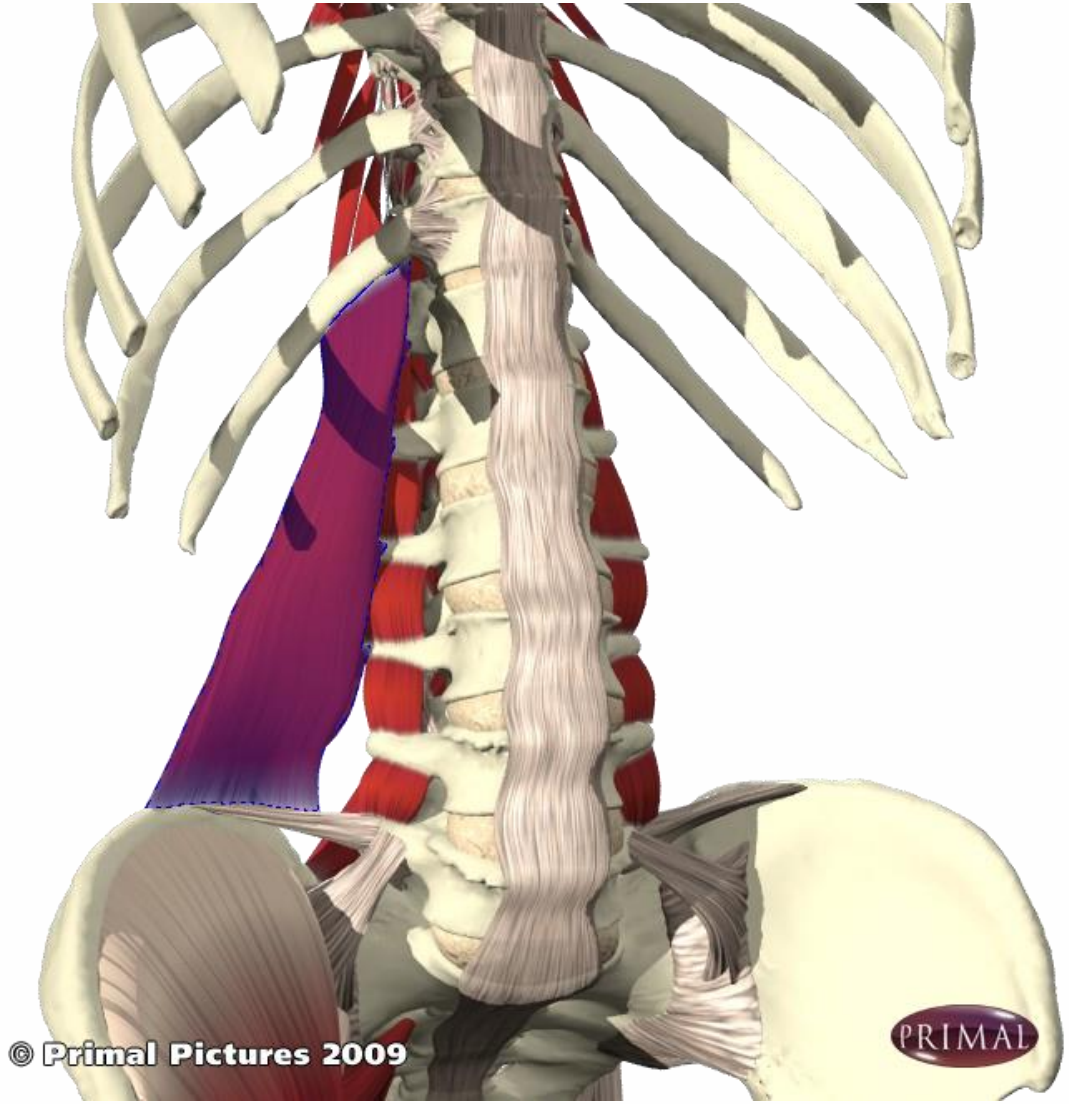
LONGISSIMUS THORACIS



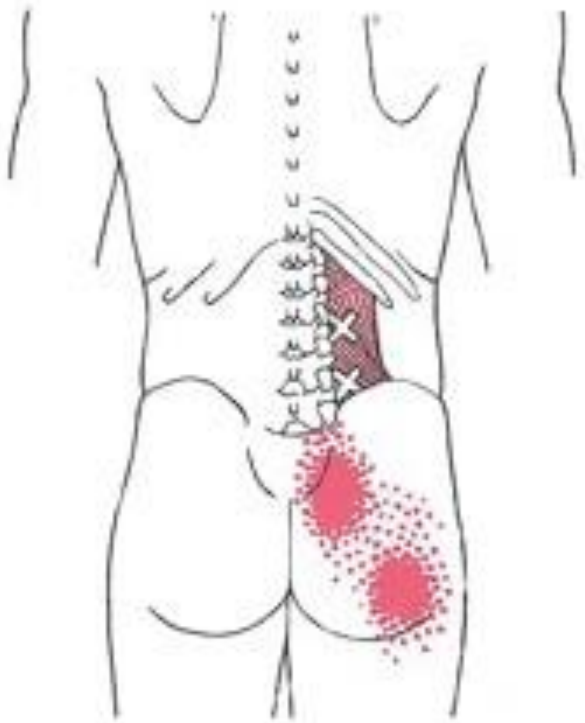


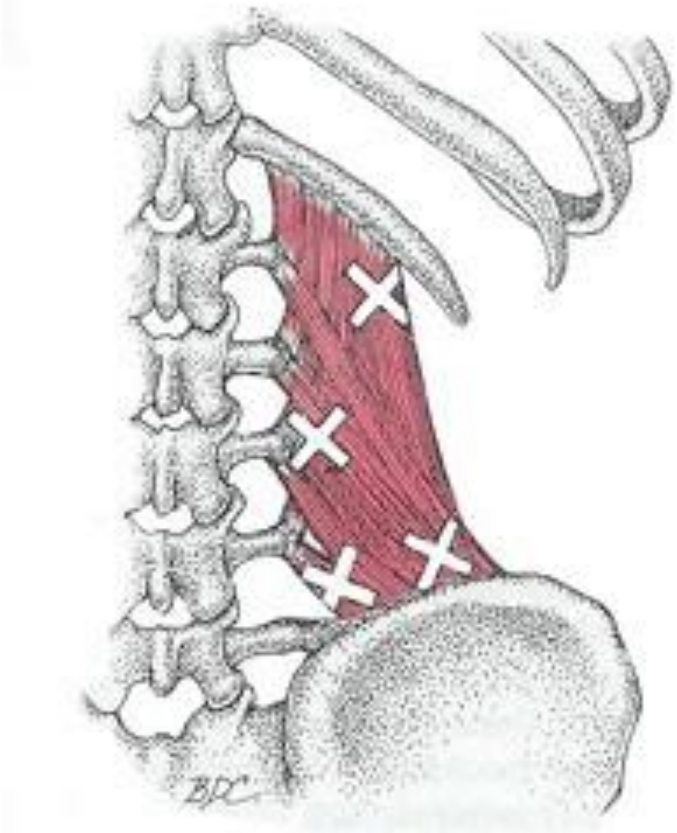
QUADRATUS LUMBORUM

- Overdominance restricts multisegmental contralateral sidebending
- Often involved in twelve rib dysfunction

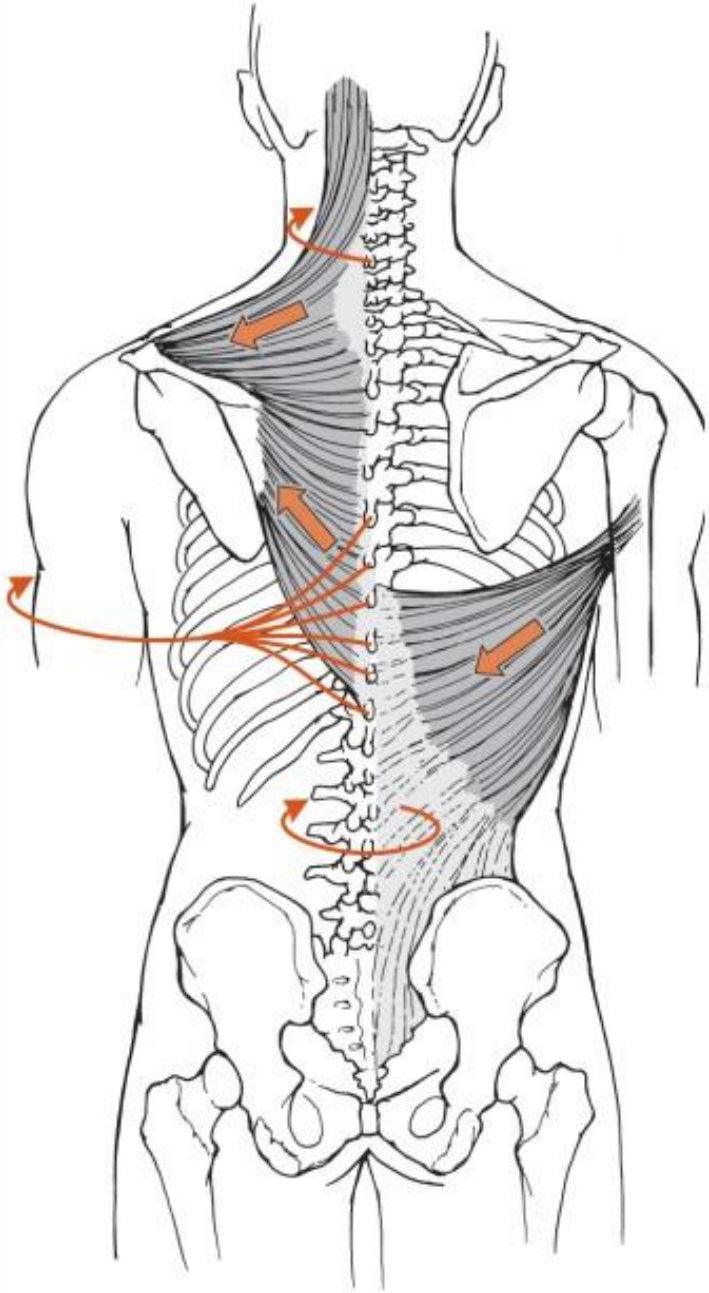


QUADRATUS LUMBORUM

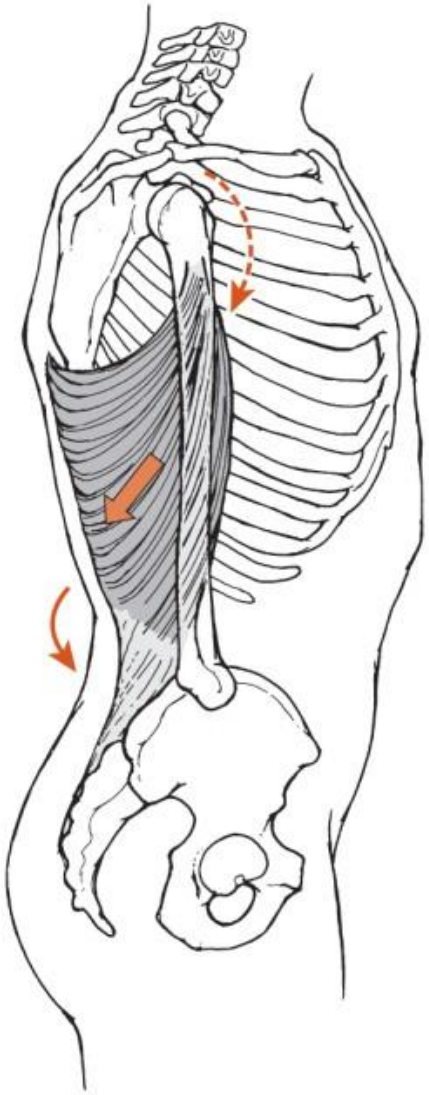




LATISSIMUS DORSI

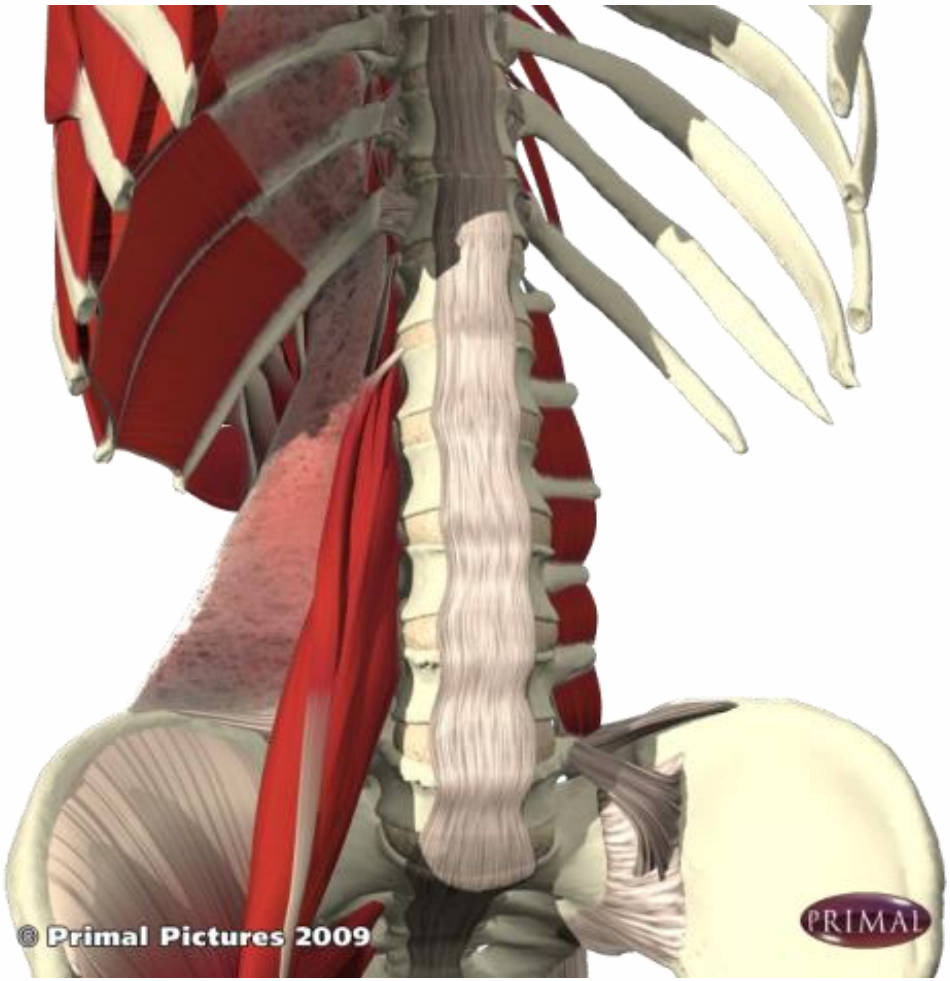


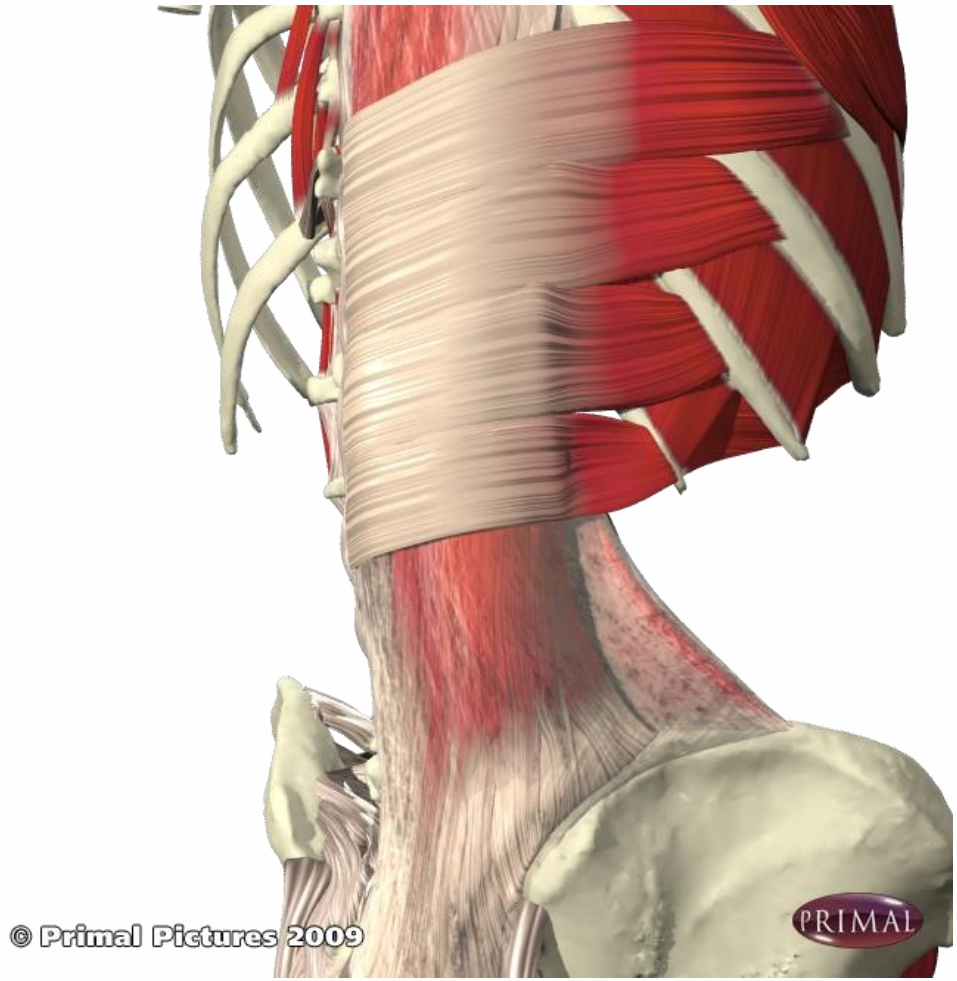
LATISSIMUS DORSI





THORACOLUMBAR FASCIA





TALKING POINTS

- Overview
- Functional anatomy
- Pathophysiology
- Evaluation
- Management

WHEN IS EXAMINATION OF THORACIC SPINE INDICATED?

- Subjective history reveals the following:
  - Complain of pain localized to the thorax or costal cage
    - Varies with respiration
    - Aggravated by twisting, pulling, pushing, or lifting, especially during overhead activities
  - History of trauma to thoracic spine or ribs

WHEN IS EXAMINATION OF THORACIC SPINE INDICATED?

- Cervical or lumbar kinematic changes:
  - Thoracic spine hypomobility is commonly associated with many cervical and lumbar spine pathophysiology
  - Stiffness in the thoracic spine increases mechanical loading on the neighboring cervical and lumbar spinal segments
  - Contributes to development of segmental instability

WHEN IS EXAMINATION OF THORACIC SPINE INDICATED?

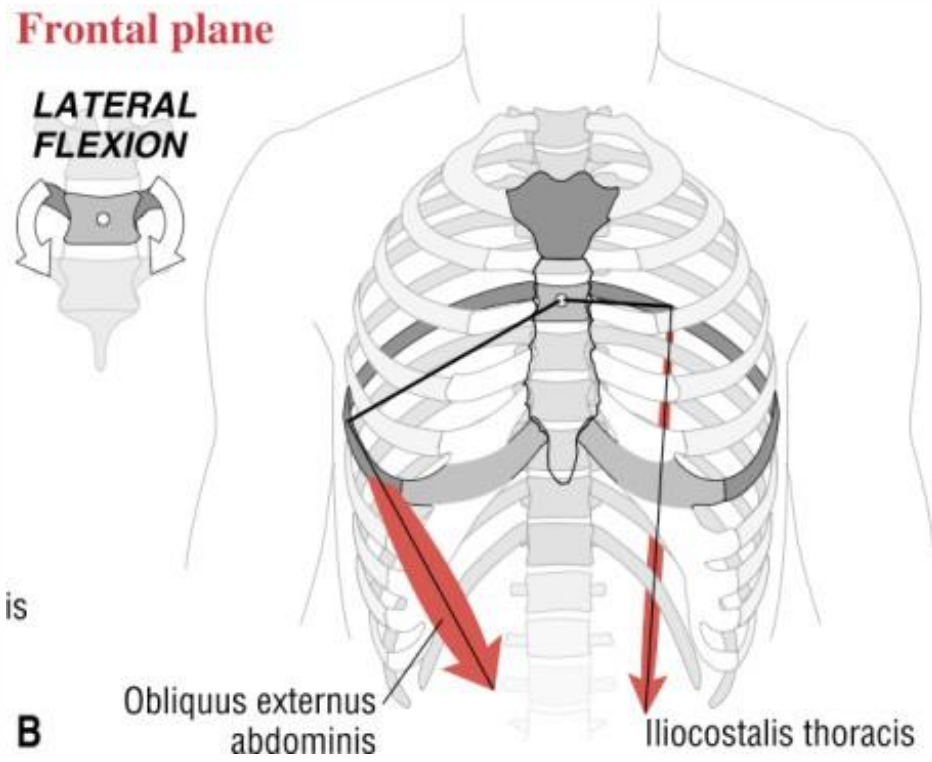
- Shoulder pathology
  - Thoracic spine hypomobility associated with impingement, rotator cuff injuries, glenohumeral joint instability
  - If limited thoracic spine mobility is noted with shoulder movements, thoracic spine exam is warranted

THE THORAX

- Pathology present?
- Are the joints mobile enough to allow respiratory excursion?
- Are the bones healthy?
- Is there passive inert stability?
- Is the motor control good enough to facilitate coordinated motion between the upper and lower quarters?
- Is the function pathological, normal, efficient?

RESPIRATORY RIB DYSFUNCTION

- Restriction in the excursion of the rib with respiration
  - Often painless
  - No palpable asymmetry in rib position
  - Primarily a soft tissue restriction that prevents full rib motion
  - Occurs in groups
- No rib cage dysfunction and therefore no restriction to breathing



STRUCTURAL RIB DYSFUNCTION

- Greater incidence of rib torsions in the lower ribs secondary to increased mobility
- Traumatic in nature
- Persists in flexion and extension
- Key rib for a group of ribs with respiratory restriction
- Painful



STRUCTURAL RIB DYSFUNCTION

- Rib torsions secondary to segmental dysfunction of facet joints
  - **External torsion**
    - Associated with an ERS lesion of segment segment
  - **Internal torsion**
    - Associated with an FRS lesion of superior segment

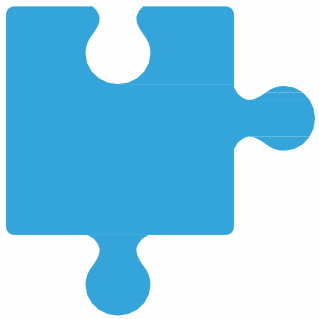


TALKING POINTS

- Overview
- Functional anatomy
- Pathophysiology
- Evaluation
- Management

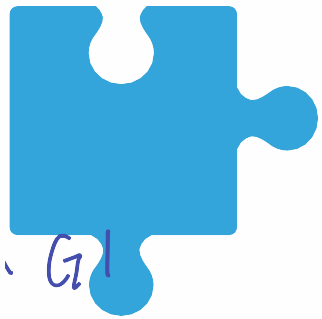
SUBJECTIVE HISTORY: DIFFERENTIAL DIAGNOSIS

- CAUTION!!! The thoracic spine is the most common site of referred pain within vertebral column
  - Thoracic pain may originate from all of the viscera
  - Systemic illnesses, such as rheumatoid arthritis, ankylosing spondylitis, malignancy, and other conditions which cause pain referral must be included in the differential diagnosis



SUBJECTIVE HISTORY: DIFFERENTIAL DIAGNOSIS

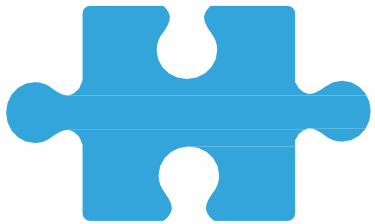
- Determine whether the pain is provoked or alleviated by the following:
  - Movement or Posture
  - Respiration
  - Eating or drinking
  - Exertion





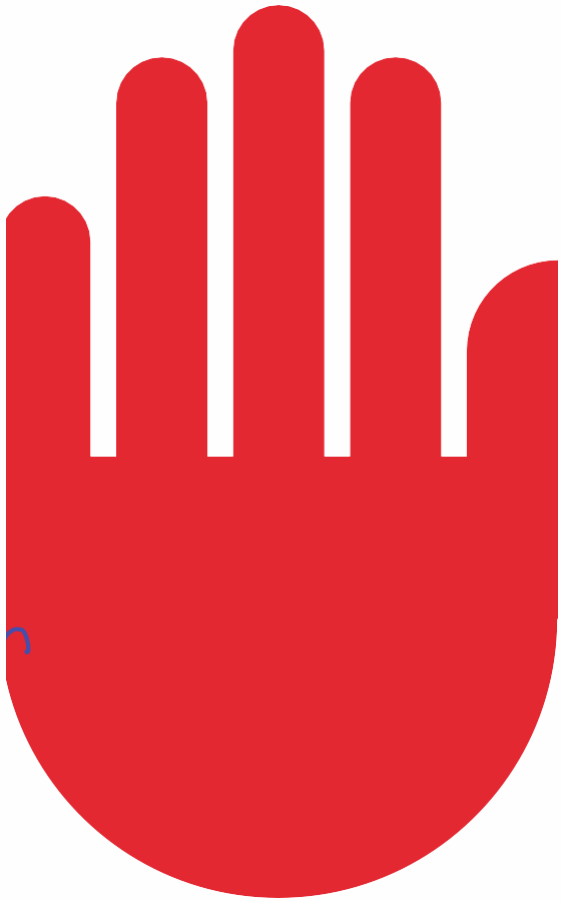
SUBJECTIVE HISTORY: SPECIAL QUESTIONS

- Does the pain occur on inhalation, exhalation, or both?
- Is the pain affected by coughing, sneezing, or straining?
- Bowel or bladder abnormalities?
- Upper and lower extremity numbness, tingling or weakness?
- Long-term steroid use? Anti-coagulants?
- Osteopenia or osteoporosis?



SUBJECTIVE HISTORY: RED FLAGS!

- Chest pain without a pattern
- Dyspnea
- Nausea or vomiting
- Night pain
- Recent of current UTI
- Dull, gnawing pain, burning



OBJECTIVE EXAMINATION: STATIC OBSERVATION

- Posture
  - Obvious swelling or wasting
  - Chest wall shape
- Quiet and deep respiration
  - Symmetry
  - Expansion



OBJECTIVE EXAMINATION: DYNAMIC OBSERVATION

- Active range of motion
- Cardinal plane motion with overpressure
  - Thoracic
    - Assess trunk motion in sitting with arms folded across chest
    - Can palpate TPs during AROM to assess segmental mobility of abnormal areas
  - Cervical
    - Rotation with ipsilateral sidebending indicative of upper thoracic dysfunction
    - Palpate cervicothoracic junction and nuchal ligament
  - Shoulder Girdles
    - Elevation, depression, protraction, retraction
    - Performed with isometric resistance

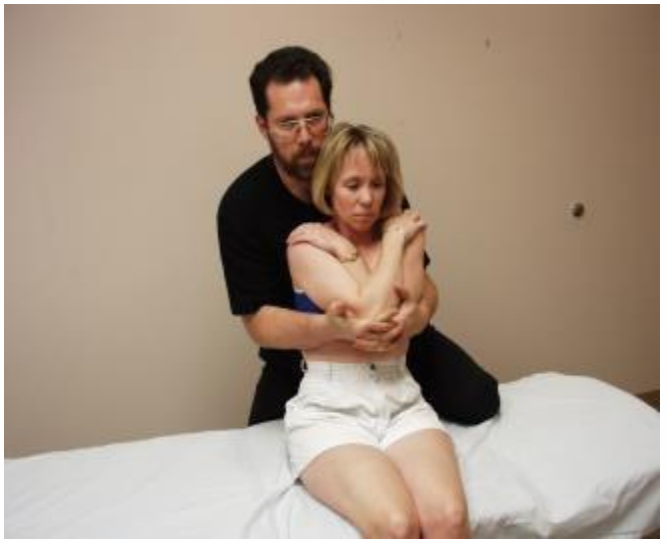
OBJECTIVE EXAMINATION: DYNAMIC OBSERVATION

- If no restriction or symptom reproduction is uncovered during cardinal plane AROM, assess combined movements as follows:
  - Rotexion
    - Full rotation then induce ipsilateral sideflexion
  - Latexion
    - Full sideflexion then induce contralateral rotation

OBJECTIVE EXAMINATION: STRESS

- Compression & distraction
  - Via head and neck for upper thoracic
  - Via shoulder girdle for middle/lower
  - Note: if symptoms reduce with distraction, then the problem is likely mechanical and will respond to physical therapy intervention



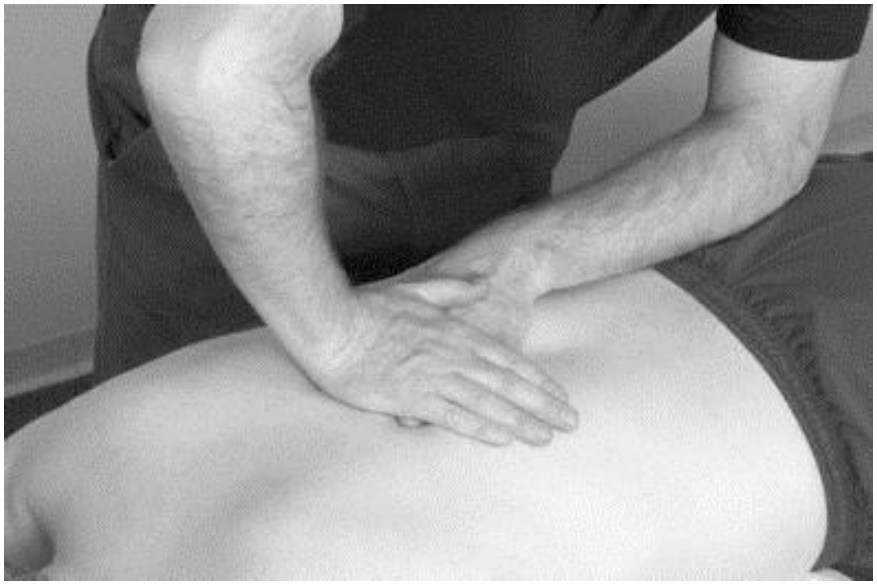


OBJECTIVE EXAMINATION: PASSIVE MOBILITY

- PAIVMS
  - PA to spinous or transverse process
  - Rib Springing, sternum
  - Specific zygopophyseal and costal joint glides
- PPIVMs
  - Assess areas identified during active mobility testing as being either stiff or too mobile
  - Palpate either interspinous space or TPs
  - Apply overpressure to dysfunctional segments to determine end-feel

PASSIVE MOBILITY TESTING

- PAIVMS
  - PA to spinous or transverse process
  - Rib springing, sternum
  - Specific zygopophyseal and costal joint glides





PASSIVE MOBILITY TESTING

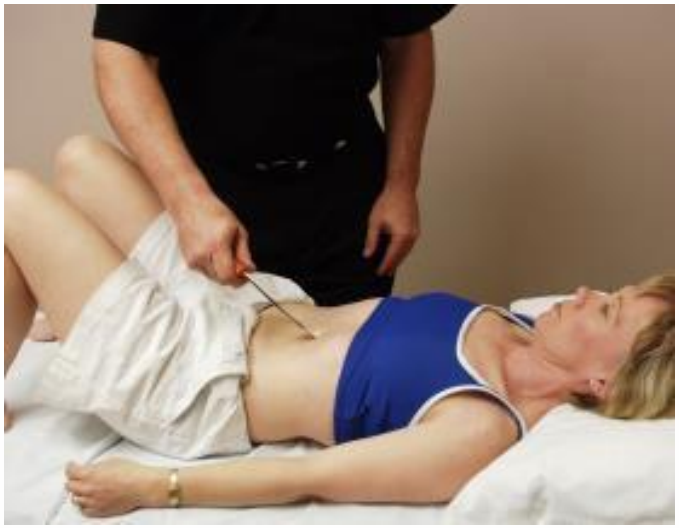
- PPIVMs
  - Assess areas identified during active mobility testing as hypo- or hypermobile
  - Palpate either interspinous space or TPs
  - Apply overpressure to dysfunctional segments to determine end feel and symptom response



OBJECTIVE EXAMINATION: NEURODYNAMICS

- Dural mobility/sensitivity tests
  - Neck flexion/slump
  - T1 Stretch or ULTT3
  - Scapular retraction
- Neurological tests
  - Lower motor neuron signs
  - Upper motor neuron signs
  - Sensory
  - Motor





OBJECTIVE EXAMINATION: CLINICAL PEARLS

- If familiar symptoms are reproduced with with a thoracic spine screen, diagnostic imaging is recommended secondary to the risks of visceral origin pathology
- If a patient complains of pain between the scapulae at or above the level of the inferior angles, it is prudent to rule out neck origin of symptoms given the C4-C7 facets refer to this region

TALKING POINTS

- Overview
- Functional anatomy
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CASE STUDY

- The patient is a 37 year old male who presents with 3 day history of left anterolateral chest pain. He describes his symptoms as a general dull ache, but with intermittent sharp sensations. The dull ache is rated a 3/10 on the visual analog scale and the sharp pain an 8/10. His symptoms are made worse by laying supine or prone, pressing a door open or pressing up from a chair with his left upper extremity, coughing, sneezing, or yawning.



MANAGEMENT THEORY

- Generate a specific problem list
- Organize the problem list into component parts
- Assign an intervention to each component
- Execute the chosen interventions
- Prescribe reproducible home program

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